A Cohort-Aware Ensemble Reinforcement Learning Framework for Personalized Glycemic Control

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

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INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that this project report titled **"A COHORT-AWARE ENSEMBLE REINFORCEMENT LEARNING FRAMEWORK FOR PERSONALIZED GLYCEMIC CONTROL"** is a bonafide record of work done by **HRUTHIK KRISHNA PAVARALA (228W1A12B4), SUDHAMSH MAGANTI (228W1A12A5)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2024-25.

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	A Cohort-Aware Ensemble Reinforcement Learning Framework for Personalized Glycemic Control
2	Student Names & Numbers	Hruthik Krishna Pavarala (228W1A12B4) Sudhamsh Maganti (228W1A1A5)
3	Name of The Guide	Dr. G. Kalyani, Associate Professor
4	Research Group	Medical Data Analysis
5	Application Area	Health care
6	Aim of the Project	To develop an advanced AI system using a cohort-aware Meta-Ensemble Deep Reinforcement Learning framework. The main goal of this project involves building an adaptive system which controls blood sugar levels for various patient populations through real-time learning of insulin delivery methods which automatically adjust to patient differences while keeping blood sugar levels stable without needing patients to announce their food intake.
7	Project Outcomes	1. A well-trained RL agent capable predicting optimal insulin dosages required based on glucose trends, physiological features, and real-time patient state. 2. Improved patient outcomes across multiple cohorts by reduced occurrences of hypogycemia and hyperglycemia.

Student Signatures

1.

2.

Signature of the Guide

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ABSTRACT

Diabetes treatment needs accurate insulin dosage because incorrect management leads to dangerous blood sugar levels which can either drop too low or rise too high. The research paper presents a new meta-ensemble reinforcement learning system which combines Soft Actor Critic (SAC) with Twin Delayed Deep Deterministic Policy Gradient (TD3) to develop personalized insulin control systems with closed-loop operation. The model uses a trainable meta-controller to merge these policies. This enables it to modify insulin doses through patient-specific physiological state information. The clinical framework development uses a multizone reward system which defines target glucose levels and establishes penalties for negative events, while supporting consistent blood sugar control. The system contains a safety-constrained control layer which enforces physiological boundaries through insulin-on-board constraints and predictive hypoglycemia detection. The simulation environment contains realistic stochastic meal scenarios which model the unpredictable nature of time and how patients react to food intake. The proposed framework is evaluated against SAC and TD3 baseline agents across child, adolescent, and adult cohorts. The experimental data show that the model produces a Time-in-Range (TIR) value of 65.99% for children and 82.91% for adolescents, and 68.72% for adults during short-term testing. The model maintains its stability in long-term testing because it achieved a Time in-Range (TIR) of 79.97% during 10-day simulations with the adolescent group. The results demonstrate that the proposed approach effectively enhances glycemic control while delivering safe insulin delivery. This suggests it could serve as a personalized decision-support system for diabetes management.

Keywords: *Reinforcement Learning , Diabetes , Closed-Loop Insulin Delivery , Meta-Ensemble Learning , SAC · TD3 , Glycemic Control*

CHAPTER-1
INTRODUCTION

1.1 Origin of the Problem

Diabetes exists as a chronic metabolic disease which affects more than seven million people throughout the globe. People who have Type 1 Diabetes need external insulin because their pancreas fails to produce this essential hormone. The management of glucose levels between 70 and 180 mg/dL requires patients to achieve perfect harmony between their insulin administration and their eating habits and exercise routines and their body's natural responses. The system keeps its balance through an extremely complex process because glucose and insulin levels shift unpredictably.

Traditional diabetes management depends on patients to follow manual steps which include checking their blood sugar and figuring out their food intake and giving themselves insulin based on these calculations. The process contains multiple stages which produce numerous mistakes because people can generate dangerous blood sugar changes through their inaccurate carbohydrate counting and their wrong choices of insulin amounts. Hypoglycemia which is low blood glucose levels leads to immediate life-threatening emergencies that include dizziness and confusion and seizures and loss of consciousness. The development of long-term health problems from hyperglycemia includes three major conditions which affect the kidneys and nerves and heart and blood vessels. The slow entry of insulin into the body generates unpredictable blood sugar changes which create dangerous situations for patients who need to take insulin.

The manual method creates a high mental load for patients because they need to stay alert and make choices at all times during their daily routine. The different ways people react to insulin along with their eating schedules and their physical activities and their stress responses create complex challenges for managing blood sugar levels. The established method of using fixed rules and standard doses fails to deliver steady blood sugar control because it needs systems which can learn from their environment.

The development of automated closed-loop insulin delivery systems which function as artificial pancreas systems solve the problems that users face with their existing systems. The systems work to merge continuous glucose monitoring (CGM) with advanced control algorithms which perform automated insulin delivery. Deep Reinforcement Learning (DRL) has become a leading solution for

these applications because it allows agents to develop their best control strategies through ongoing environmental interactions. The reinforcement learning system obtains its knowledge from reward signals instead of using labeled data which makes it suitable for managing intricate sequential physiological systems with unpredictable behavior.

In this project, SimGlucose simulation environment serves as a tool to create authentic glucose-insulin system models which help the reinforcement learning system learn insulin dosage evaluation through safe methods. The system functions independently from meal notifications because it provides complete insulin management through its automatic system which reacts to current body conditions. The proposed framework achieves stable glucose control through its combination of ensemble learning with safety-based control systems and clinical knowledge-based reward systems. The main issue arises from the requirement to develop an automated diabetes management system which minimizes human errors while providing personalized care for each patient. The project uses reinforcement learning with simulation training to build an artificial pancreas system which provides better blood sugar control while minimizing dangerous medical situations and improving life standards for people who have diabetes.

1.2 Basic definitions and Background

1.2.1 Glucose–Insulin Dynamics:

The human body shows glucose–insulin dynamics through its natural connection between blood glucose levels and insulin hormone function. People who have Type 1 Diabetes need to take insulin from outside their body because their pancreas stops making this essential hormone. The body experiences rising blood glucose levels after eating because carbohydrates enter the bloodstream but insulin helps cells absorb glucose which leads to blood glucose reduction.

The system shows nonlinear behavior because various elements including meal schedules and insulin processing and exercise patterns and personal body chemistry affect its operation. The slow response of insulin after injection creates problems for doctors when they try to forecast what will happen to blood sugar levels. The system contains multiple difficult elements which make the control of glucose levels challenging.

The design of automated insulin delivery systems requires complete understanding of these system dynamics.The SimGlucose simulator serves as the mathematical model for this

project which lets reinforcement learning agents develop their best insulin administration methods through simulated practice.

1.2.2 Continuous Glucose Monitoring (CGM):

Continuous Glucose Monitoring (CGM) systems function as wearable devices which deliver ongoing blood glucose level measurements through scheduled monitoring sessions. Continuous tracking through CGMs provides users with better glucose pattern visibility which traditional finger-prick methods do not offer.

The CGM system records glucose values together with time-based data which shows how glucose levels change during rising and falling periods. The data serves as an essential resource to estimate upcoming blood sugar readings while helping users determine when to take their next insulin dose. The system design needs to handle CGM measurement interference together with delayed sensor feedback.

The project requires CGM data as the main source which reinforcement learning agents need to function. The system analyzes these readings together with calculated features which include rate of change to produce intelligent and adaptive insulin dosing decisions.

1.2.3 Closed-Loop Insulin Delivery Systems:

The artificial pancreas system operates as a closed-loop insulin delivery system which uses CGM data to run control algorithms that manage insulin administration. These systems actively track blood sugar levels while they handle insulin delivery through their automated system which needs no human input.

The purpose of a closed-loop system exists to keep blood sugar levels at safe points while it works to prevent both low blood sugar and high blood sugar episodes. Closed-loop systems offer better glycemic control through their design which reduces patient responsibilities when compared to traditional open-loop systems.

The project achieves its goal through reinforcement learning which creates a system that operates without any external control. The agent operates independently within the simulated environment to select insulin doses based on its current physiological state.

1.2.4 Reinforcement Learning:

The field of machine learning known as Reinforcement Learning (RL) allows agents to develop optimal decision-making methods through their ongoing interaction with their surrounding environment which aims to achieve maximum reward accumulation across

extended periods. The RL system operates independently from supervised learning because it learns through reward and penalty feedback rather than using pre-labeled training data. The system operates perfectly for sequential decision problems because its delayed action results in long-term effects which makes it suitable for medical applications like diabetes treatment with insulin.

RL functions at its best when it needs to control systems which show both complex behavior and continuous changes through their adaptive control requirements. The agent in personalized insulin delivery systems learns about patient body responses to various insulin doses which occur during different meal times[11] and glucose level changes and natural body changes. The agent uses SimGlucose to test its strategies because the simulation environment protects actual patients from any potential harm which results from experimental mistakes.

The learning system requires students to track their current body condition before they must decide their insulin dosing and they receive feedback about their blood sugar control performance. The agent learns to improve its decision-making process through experience which shows which choices produce better results. The reward function supports glucose level control within safe zones while it creates penalties for patients who develop hypoglycemia or hyperglycemia.

1.2.5 Exploration vs Exploitation in RL:

Reinforcement learning depends on its two fundamental elements which consist of exploration and exploitation to teach agents about optimal behavior. Exploration refers to the process of trying new or uncertain actions to discover potentially better strategies, while exploitation focuses on selecting actions that are already known to produce high rewards. The practice of achieving correct operational balance between these two elements creates the path to achieve learning success and operational success.

The healthcare system requires its staff to keep proper balance because any dangerous action will cause major harm to patients who need insulin treatment. The process of excessive exploration leads to dangerous insulin dosage changes which create unstable situations that threaten patients with severe blood sugar level changes. The system faces problems when it uses too much exploitation because it sticks to a few fixed approaches which block its ability to adjust to changing physiological situations including different patient meal consumption

and body reactions.

The Soft Actor-Critic (SAC) agent leads the exploratory process in this project through its stochastic policy which produces multiple adaptive insulin dosage decisions. The system achieves stability through this approach because it manages both unpredictable blood sugar levels and fast-moving changes in glucose levels. The Twin Delayed DDPG (TD3) agent supports exploitation through its deterministic policy[15] which delivers reliable and predictable insulin administration.

1.2.6 Safety in Reinforcement Learning:

The application of learning-based methods to actual systems requires Safety in Reinforcement Learning (RL) because these systems need protection during their operation. Standard RL algorithms learn through trial and error, which may lead to unsafe or harmful actions during the exploration phase. The learning process needs safety mechanisms because diabetes patients who receive wrong insulin doses will develop dangerous blood sugar levels which lead to hypoglycemia and hyperglycemia.

The solution needs safety constraints which will establish boundaries for agent actions to keep them within medical safety parameters. The system uses these restrictions to protect itself from dangerous choices which emerge during its learning process. The system achieves safety through three methods which include rule-based checks and action clipping and predictive mechanisms that forecast action effects before they occur. The system protects patients from harm through its design yet it allows the agent to keep learning without creating any safety problems.

The project includes a Safety Layer which performs continuous monitoring to control how insulin doses get prescribed. The safety layer performs three essential functions to protect patients because it stops insulin distribution during low blood sugar levels and it tracks insulin delivery to prevent dangerous buildup and it decreases insulin amounts when blood sugar levels show quick decreases. The system contains a predictive component which forecasts upcoming blood sugar levels while it stops insulin delivery when it detects potential danger.The safety measures operate to protect the reinforcement learning agent while it learns effective control methods through safe operating areas. The system achieves performance-reliability balance through its combination of safety constraints with adaptive learning which makes it suitable for actual closed-loop insulin delivery systems.

1.2.7 SAC AGENT:

The Soft Actor-Critic (SAC) operates as an off-policy reinforcement learning algorithm and serves as one of the core base agents for insulin control in this project. The system operates to reach its maximum expected reward while keeping policy entropy at its peak because this approach enables efficient exploration together with stable learning in continuous action spaces. The SAC system operates through continuous control which makes it an ideal choice for insulin dosing because it needs exact operational commands.

The SAC system generates multiple action outputs through its stochastic policy which enables it to handle various physiological situations including meal disruptions and fast-changing glucose levels. The model functions through an actor-critic system which contains several critic networks that improve learning stability while minimizing the problem of prediction errors becoming too high. The agent continues to search for different methods because the entropy-based objective function prevents it from getting stuck in basic solution paths.

The project uses SAC to generate smooth exploratory insulin dosing decisions which help during uncertain glucose levels that change quickly. The system merges its output together with TD3 agent results through a meta-controller which creates a system that maintains both exploration and stability. The system performance reaches higher levels through this integration which produces better glucose management results for various patient conditions.

1.2.8 TD3 AGENT:

The Twin Delayed Deep Deterministic Policy Gradient (TD3) functions as an off-policy reinforcement learning system which serves as a primary base agent for managing insulin dosage in this project. The system includes twin critic networks which provide stability solutions for actor-critic methods by decreasing overestimation errors while producing better value estimation results. The system updates its policy at delayed intervals which helps the actor network to learn with better stability and produces more dependable learning results.

The deterministic policy which TD3 follows produces exact action outputs through its consistent method. The system operates through a method which works best for situations that need constant control to maintain glucose levels at their normal range. The system uses

target policy smoothing which introduces minor disturbances to training actions for better system stability and to stop sudden insulin dose changes.

The project depends on TD3 to deliver dependable insulin dosage selections which help prevent random glucose level changes from occurring. The system combines SAC agent exploratory actions with output data through a meta-controller which helps the system achieve both stability and adaptability. The system achieves better performance through its combination of TD3 and SAC because TD3 provides reliable results while SAC lets the system explore new possibilities which leads to safer and better glucose control.

1.2.9 META ENSEMBLE AGENT:

The Meta Ensemble Agent uses two reinforcement learning agents Soft Actor-Critic (SAC) and Twin Delayed DDPG (TD3) to produce optimal insulin dose recommendations through its integrated system. The system operates through a trainable meta-controller which merges the output of two agents during live operational phases. The meta-controller develops state-based weighting systems which allow it to control how each agent contributes during different physical states including increasing glucose levels and stable glucose patterns.

The ensemble approach combines the strengths of both algorithms to create an effective system. The SAC system produces adaptive actions which allow exploratory behavior when glucose levels become uncertain during postprandial spikes. The TD3 system delivers stable control which proves vital for precise insulin delivery during steady-state operations. The meta-controller bases its decision-making process on these combined behaviors to produce steady decisions which adapt to different situations.The meta-controller operates through a neural network which processes state information after normalization to generate weighting values that control SAC and TD3 action selection. The system learns weight values through its training process which uses environmental feedback to determine which agent shows better performance under particular environmental conditions. The learning method enables the ensemble to develop its performance through self-improvement during each learning cycle.The ensemble agent functions through a unified replay storage system which collects environmental data to support simultaneous training of SAC and TD3. The agents learn together through this system which helps them work better as a team while they learn more from fewer experiences. The safety layer receives the ensemble's combined action to apply clinical restrictions which protect patients from hypoglycemia and control their insulin-on-

board (IOB) levels and maintain proper insulin administration.

The Meta Ensemble Agent develops better system strength and flexibility and safety performance when compared to working with separate agents. The system stabilizes blood sugar levels while extending diabetes patients' blood glucose control duration and decreasing their risk of developing dangerous medical events. The ensemble approach solves automated insulin control problems in closed-loop diabetes management systems through its dual system which combines exploration with stability to deliver dependable results.

1.3 Problem Statement:

Traditional diabetes management requires patients to measure their carbohydrate intake correctly while they must follow precise schedules for their insulin injections. The process of estimating carbohydrate intake and scheduling insulin injections produces minor mistakes which result in major blood sugar level fluctuations that lead to dangerous low and high blood sugar emergencies. Medical professionals face obstacles because they must evaluate patient dosage requirements through manual assessment which produces insufficient decision-making results.

This project addresses this challenge through Reinforcement learning (RL) which enables an autonomous agent to develop optimal insulin delivery methods by studying individual patient glucose–insulin response patterns. The RL agent learns from body signals instead of human input about carbohydrates because it tracks blood sugar patterns and insulin effects on the body to develop its own treatment strategy for keeping blood sugar levels safe. The data-based system shows promise to reduce patient mental demands while it improves blood sugar management through personalized glucose monitoring which delivers better results than conventional approaches.

1.3.1 Objectives

1. To establish a training environment based on SimGlucose which will simulate conditions for reinforcement learning agent development.
2. To build a meta-ensemble reinforcement learning system which will unite SAC and TD3 algorithms for insulin control operations
3. To develop an independent control system which obtains its best insulin dosage methods by learning through environmental interactions.
4. To include body weight as a physiological characteristic which will help it create personalized decision-making models.

5. The clinical reward function needs to maximize Time in Range (TIR) while it works to reduce both hypoglycemia and hyperglycemia occurrences.
6. The control system needs to establish safety boundaries which block unsafe insulin delivery methods while maintaining clinical trust in the system.
7. We need to test our framework through patient groups which include adults and adolescents and children who face authentic meal situations.

1.3.2 Outcomes

1. The development of a strong meta-ensemble reinforcement learning system achieved success through its application in closed-loop insulin management.
2. The framework which has been proposed delivers better glycemic outcomes through its TIR results which reached 82.91% for adolescents and 68.72% for adults and 65.99% for children.
3. The system proved its ability to reduce both hypoglycemic and hyperglycemic occurrences throughout all patient groups which were monitored.
4. The model achieved its goal of modeling glucose-insulin interactions while it modified insulin dosage through actual body responses which occurred at that time.
5. The system achieved better decision-making abilities because it combined physiological features which included body weight to generate personalized results.
6. The safety layer provided steady insulin delivery which protected patients from dangerous hypoglycemic episodes in almost all situations.
7. The developed system operates through a closed-loop system which does not require meal announcements to function and it shows strong potential for actual clinical use.

1.4 Societal Impact of Proposed Work:

1. The automated insulin delivery policy improves clinical decision-making through its system which provides instant tailored insulin recommendations that reduce the mental work required from healthcare providers.
2. The system achieves dependable results through its strong ensemble RL agent which includes a built-in Safety Layer that creates predictable outcomes for clinical trust and patient treatment management.
3. The correct and steady insulin administration reduces the frequency of hypoglycemia and hyperglycemia events. This results in better patient safety, improved glycemic

management, and longer-term health benefits.

4. The system enables continuous monitoring and adaptive learning, allowing it to improve performance over time as more patient data becomes available.
5. It reduces dependency on frequent hospital visits by supporting remote and at-home diabetes management, which is especially beneficial for patients in rural or underserved areas.
6. The framework encourages data-driven healthcare practices by utilizing real-time physiological data to make informed treatment decisions.

CHAPTER-2
REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and

2.1 Description of Existing Systems

[1] Integrating Neural Differential Forecasting with Safe Reinforcement Learning for Blood Glucose Regulation

Authors: Yushen Liu , Yanfu Zhang

Year : 2025

Observations:

This literature work presents a comprehensive study an in-depth analysis of advanced Artificial Intelligence methods which include Reinforcement Learning and Neural Differential Models to enhance automated blood glucose control for Type 1 Diabetes patients. The paper introduces a new system which unites adaptive learning techniques with predictive modeling to solve the problems of insulin dosing when dealing with unpredictable body responses during meals and individual patient characteristics. The system employs a Neural Ordinary Differential Equation (NeuralODE) model to forecast upcoming glucose levels but the reinforcement learning-based controller selects the best insulin doses. The system predicts upcoming glucose levels to deliver individualized treatment which enhances patient outcomes through its forward-thinking glucose management system.

A key contribution of the study is the inclusion of a probabilistic safety mechanism that evaluates the risk of each insulin decision before execution. The model goes beyond standard methods which mainly focus on glucose control because it stops dangerous medical errors which include giving too much insulin and making quick insulin adjustments that result in low blood sugar. The system protects patient safety through its safety restriction system which stops dangerous actions from happening while its uncertainty evaluation system makes decisions to cancel or change dangerous actions. The experimental results demonstrate that patients achieved better time-in-range glucose control while experiencing fewer hypoglycemic episodes when compared to their baseline treatment methods. The study shows that real-world usefulness depends on having reliable physiological data together with unbroken monitoring systems. The study shows that predictive modeling together with safety-aware reinforcement learning systems will become an effective method to build diabetes

treatment systems which adapt to individual needs and focus on patient care.

[2] An Improved Strategy for Blood Glucose Control Using Multi-Step Deep Reinforcement Learning

Authors: Senquan Wang , Weiwei Gu

Year : 2024

Observations:

This literature work presents a Deep Reinforcement Learning system which operates at an advanced level to help Type 1 Diabetes patients achieve better blood glucose regulation through enhanced insulin delivery methods. The paper presents a new Multi-step Deep Q-Network (MDBG) framework which enhances standard reinforcement learning techniques by modeling how insulin produces delayed biological effects. The system models glucose control through an exponential decay system which converts the problem into a Markov Decision Process to show how previous insulin injections affect current blood sugar measurements. The model learns from crucial experiences through multi-step learning and prioritized experience replay which enables it to reach its goals faster while making better choices. The system applies reinforcement learning to pursue reward goals which generate adaptable glucose control results that surpass basic immediate responses.

The research achieved its main goal by demonstrating that the proposed model produces better glycemic control results than conventional methods. The MDBG algorithm maintains blood glucose levels within target range for longer periods while it decreases the occurrence of severe hypoglycemia and hyperglycemia events. The model shows improved glucose sensitivity because it reacts better to glucose input changes through its system which delivers small insulin doses at multiple times during the day. The research shows two main problems because it uses virtual testing spaces and it does not properly assess how people behave in their normal environment through their eating habits and their exercise routines and their individual differences. The authors showed that multi-step deep reinforcement learning methods would work well for diabetes management but they need to test their system with actual clinical data to prove its effectiveness in medical environments.

[3] Fully-Automated Patient-Agnostic Diabetes Management with Deep Reinforcement Learning

Authors: Thomas Milton , Robert Lieck

Year : 2023

Observations:

This research introduces an original method for automatic diabetes treatment which applies Deep Reinforcement Learning to create a complete patient-independent system that manages Type 1 Diabetes. The paper presents a Soft Actor-Critic (SAC) model which trains on an improved version of the UVA/Padova simulator that uses actual meal data to boost its accuracy and its ability to produce consistent results. The system operates without human direction because it learns to recognize different patients through its automated process which replaces standard techniques that need users to report their meals and adjust system settings. The model decides insulin delivery through its state space which contains physiological data about patients including their age and their body measurements and their entire insulin usage during the day. The reinforcement learning framework allows the system to develop perfect insulin administration methods which keep blood sugar levels at their proper range through its process of maximizing long-term rewards.

The study presents a vital contribution because it manages to reach excellent glucose control results which simultaneously work well with new patients who were not part of the training data. The proposed model achieves its highest time-in-range (TIR) value at 80.3% which exceeds the performance of traditional PID controllers and other current advanced methods. The system learns to deliver insulin safely through its adaptive dosing approach which helps it prevent both low and high blood sugar episodes. The research team identified several study limitations which stem from their use of artificial testing environments and their restricted patient population and their failure to include actual physical exercise in their study. The authors propose three main directions for future work which will include diverse data sources and real-time system monitoring and clinical evaluation of the system. The research paper proves that patient-agnostic reinforcement learning models enable diabetes management systems to reach their full potential for automated safe and efficient operation.

[4] Reinforcement Learning: A Paradigm Shift in Personalized Blood Glucose Management for Diabetes

Authors: Lehel Dénes-Fazakas , László Szilágyi

Year : 2024

Observations:

This research establishes a diabetic control system which uses Artificial Intelligence through Reinforcement Learning to optimize insulin delivery for patients who have Type 1 Diabetes. The

research team works to create an automated system which learns optimal insulin dosing methods through its own learning process without requiring ongoing human supervision. The system operates through sequential decision-making to model glucose-insulin interactions which allows it to adapt to changing physiological states while predicting suitable insulin dosages. The method uses individual patient details together with blood sugar changes to create tailored treatment plans which help patients reach better glucose control through precise methods.

A key observation from the study is that the proposed model significantly enhances glucose control performance by maintaining higher time-in-range levels and reducing the occurrence of hypoglycemia and hyperglycemia. The system shows excellent ability to adjust and maintain stability when it faces unexpected changes in real-world diabetes control situations. The research includes particular restrictions because it depends on computer-generated test areas and fails to connect with actual daily habits which people follow through their eating patterns and exercise routines. The authors established that AI-driven insulin management systems needed more clinical testing together with actual patient data to confirm their operational ability for healthcare facility use

The research emphasizes the need for automated insulin delivery systems to achieve scalable performance which must work effectively in actual medical environments. The framework enables ongoing learning which helps the model adapt to new information for better results through time because of expanding patient data collections. The system's capability enables it to handle diabetes management over extended periods because it can adjust to the different ways patients respond to their bodies during stress and sickness and hormonal shifts and changes in their daily routines. The authors propose that AI-based systems should merge with wearable sensors and continuous glucose monitoring devices to create an improved system which provides automatic diabetes management through a fully closed-loop system. The research demonstrates how intelligent automation technology will enhance traditional diabetes treatment through its ability to develop proactive healthcare solutions which use data-based personalized treatment methods. This approach highlights the transformative potential of intelligent, adaptive systems in advancing precision medicine and improving long-term patient outcomes.

[5] A safe-enhanced fully closed-loop artificial pancreas controller based on deep reinforcement learning

Authors: YanFengZhao , YiqiTew

Year : 2025

Observations:

The research study develops a secure fully automated artificial pancreas system which operates as a complete closed-loop system through deep reinforcement learning to deliver insulin automatically for managing Type 1 Diabetes. The research presents a better version of Proximal Policy Optimization (PPO) which combines various optimization methods to achieve better training stability and efficiency. The proposed controller functions independently from traditional hybrid systems because it does not need external inputs for meal announcements because it uses continuous glucose monitoring data to calculate insulin requirements. The research team developed a dual safety system which combines reward function design for forward direction with safety adjustment module to reactively modify system behavior which delivers safe insulin treatment during fluctuating bodily states. The model learns optimal glucose control methods through its interactions with virtual patients within the Simglucose simulator which serves as the evaluation environment.

The results demonstrate significant improvement in glucose control performance, achieving a median Time-in-Range (TIR) of 87.45% while effectively reducing both hyperglycemia and hypoglycemia events. The system shows complete prevention of severe hypoglycemia and treatment failures which proves its safety performance outperforms both traditional PID systems and current DRL-based approaches. The research depends on simulated testing environments which prevents it from obtaining authentic clinical proof from medical facilities. The research faces three main difficulties because scientists cannot read their models and their training processes take too long and their treatment approaches fail to work properly for individual patients. The research demonstrates that safety systems which work together with DRL technology will create better artificial pancreas systems which operate automatically with perfect reliability also demonstrates strong robustness to meal disturbances and sensor noise, indicating its ability to operate effectively under real-world uncertainties. The enhanced PPO algorithm, combined with multiple training "tricks," significantly improves convergence compared to standard implementations. Additionally, the controller achieves personalized glucose regulation without explicitly modeling patient-specific physiology, making it more scalable. However, the reliance on simulator-trained policies raises concerns about transferability to real patients and highlights the need for further clinical testing and real-time deployment validation. The research establishes better model explanation requirements which need to undergo training simplification to achieve medical practice suitability for clinical environments.

[6] Privacy-Preserving Glycemic Management in Type 1 Diabetes: Development and Validation of a Multiobjective Federated Reinforcement Learning Framework

Authors: Juan Li

Year : 2025

Observations:

The literature text shows a privacy-protecting Deep Reinforcement Learning system which uses smart insulin delivery to help Type 1 Diabetes patients achieve better blood sugar control. The research presents a new Federated Reinforcement Learning (FRL) system which allows multiple agents to develop individual insulin dosing strategies through decentralized patient information management. The system operates with a multi-objective reward structure which strengthens Time in Range (TIR) performance and reduces both hypoglycemia and hyperglycemia and insulin consumption to achieve better clinical results than standard single-objective models.

The framework solves glucose control by using a sequential decision-making model through a partially observable Markov decision process which needs real-time glucose data for its uncertain decision-making process. The local reinforcement learning agent (SAC-based) trains on each patient device while federated learning enables model updates to reach a central server which protects patient privacy and enhances model performance for different patient groups. The system uses entropy-based reward shaping to improve its exploration capabilities which leads to better glucose level stability.

The research findings show that the proposed model delivers excellent blood sugar management through extended TIR periods while preventing any occurrence of hypoglycemia in all examined patient populations. The system produces better stability and lower glucose fluctuations when compared to present-day operational systems. However, the study has certain limitations, as it relies on simulated data from the UVA/Padova simulator and requires real-world clinical validation. The system faces two major deployment challenges because it needs to handle substantial communication expenses and deal with various patient needs. The study demonstrates that federated learning together with multi-objective reinforcement learning techniques develop diabetes management systems which maintain safety and adaptability and protect patient data privacy. The framework also highlights the importance of scalable architectures that can efficiently handle distributed patient data while maintaining consistent performance across populations.

[7] Deep Reinforcement Learning Control of Type-1 Diabetes with Cross-Patient Generalization

Authors: Mohab M.H , Federico

Year : 2020

Observations:

The literature work presents a control system which uses Deep Reinforcement Learning to enhance blood glucose management for patients who have Type 1 Diabetes. The study proposes the use of a Deep Deterministic Policy Gradient (DDPG) algorithm to develop an automated insulin delivery controller capable of handling continuous action spaces. The system models glucose-insulin dynamics through the SimGlucose environment which employs the FDA-approved UVa/Padova simulator and uses Markov Decision Process to solve sequential decision problems. The agent learns from continuous glucose monitoring (CGM) data and utilizes both current glucose values and their rate of change to make informed insulin dosing decisions.

The model functions across different patient profiles instead of focusing on a specific single patient. The system becomes better at handling physiological differences because it learns from virtual patients who represent adults and adolescents and children. The DDPG algorithm uses an actor-critic structure which combines experience replay with target networks to achieve stable learning and optimize policy selection effectively. The reward system operates to maintain glucose levels between 70–180 mg/dL which results in penalties for patients who experience either hypoglycemia or hyperglycemia.

The research findings demonstrate that the RL-based controller system performs better than conventional basal-bolus therapy because it maintains blood glucose levels within the target range for extended periods and reduces the occurrence of hyperglycemic events. The system demonstrates better generalization abilities because it operates multiple patient cases through a single trained model. The research study faces two main restrictions because it depends on artificial testing environments and it does not include validation through actual medical practice. The research fails to include complete data about how people live their lives through their eating habits and their movement patterns which creates doubts about its usefulness for actual human environments. The research demonstrates that deep reinforcement learning can create scalable insulin control systems which show promise for future development.

[8] A Hybrid Closed-Loop Blood Glucose Control Algorithm with a Safety Limiter Based on Deep Reinforcement Learning and Model Predictive Control

Authors: Shanyong Huang, Yusheng Fu

Year : 2026

Observations:

The research presents a sophisticated control system which merges Deep Reinforcement Learning with Model Predictive Control to provide safe blood sugar management for patients who have Type 1 Diabetes. The research solves the fundamental problem which is vital to protect patients during DRL-based insulin delivery system testing because reinforcement learning systems use trial-and-error methods which create dangerous situations for hospital patients. The system achieves its limitation solution through a design which merges predictive MPC functionality with a safety system that limits DRL agent operations to medical safety boundaries. The system uses real-time glucose and insulin interaction data to train a prediction model which removes the need for pre-set physiological models and allows the system to learn from data-based control systems.

The framework enables the DRL agent to connect with a simulated patient environment while an extra prediction model forecasts upcoming blood glucose levels based on present conditions and chosen actions. The MPC module runs multiple step simulations to test different actions and it removes dangerous paths which would lead to blood sugar levels going too low or too high. The safety controller operates as an emergency system which modifies insulin administration during critical events to protect patients from dangerous low and high blood sugar episodes. The system achieves superior reliability through its two safety systems which preserve the reinforcement learning system's adaptive abilities.

[9] Residual Hybrid Deep Learning Approach for Personalized Insulin Dose Estimation

Authors: Sudhamsh , Hruthik , Kalyani

Year : 2026

Observations:

This research presents a Residual Hybrid Deep Learning system which predicts individual insulin doses through its advanced framework to solve the problems that occur with standard rule-based and doctor-dependent insulin treatment approaches. The research team analyzed actual medical records from the AIM94 dataset which they converted from its original random event structure into organized time-based data for deep learning analysis. The system performs complete data

preprocessing through its cleaning process and normalization stage and missing value treatment and event encoding for physiological data which includes meals and hypoglycemia and physical exercise. The system applies cyclical coding to primary time-related data which produces better results when it works with specialized feature generation methods that create features from past glucose and insulin readings and rolling statistics and combined variables to detect short-term changes and long-term physiological patterns in patients. The model architecture combines LSTM and GRU layers which use residual connections to create an efficient information network between layers that enhances their ability to learn complex glucose–insulin relationships. The system design uses a combined approach which unites memory capacity with computational speed to create a powerful sequence processing system that predicts insulin levels correctly.

The experimental results demonstrate that the proposed model significantly outperforms traditional and baseline deep learning approaches, achieving higher predictive accuracy and better generalization across diverse patient profiles. The combination of residual learning with hybrid recurrent architectures produces better performance results because it leads to a major rise in R^2 score and decreases both MAE and MSE values. The model shows excellent performance in predicting insulin values because it maintains tight control over insulin needs while handling regular changes in daily insulin requirements. The research shows strong aspects yet it contains specific restrictions because it depends on past records and lacks instant response systems which prevent its use in complete automated insulin delivery technology. The model fails to include external lifestyle elements which consist of stress levels and dietary choices and physical activity intensity so its real-world prediction accuracy becomes uncertain.The research shows that healthcare systems need intelligent data-based solutions to manage diabetes effectively. Deep learning models function as dependable decision support systems for medical staff and their patients. The proposed system design allows for future expansion through additional multimodal data sources which will help produce improved context-based predictions. The research needs to develop reinforcement learning systems which will operate as closed-loop insulin controllers. The research needs to develop new methods which will help medical staff understand model decisions better. The system needs to prove its working ability through actual patient data which should be monitored in real healthcare settings. The healthcare delivery of personalized diabetes management needs adaptive AI systems which must connect to real-time patient monitoring devices. The development of future healthcare solutions requires better model strength and direct clinical application for systems which deliver dependable and easily expandable solutions.

[10] Toward a Fully Automated Artificial Pancreas System Using a Bioinspired Reinforcement Learning Design: In Silico Validation

Authors: Seunghyun Lee , Jiwon Kim

Year : 2022

Observations:

The research presents an artificial pancreas system which operates without human supervision through a Deep Reinforcement Learning system that draws inspiration from biological systems to reproduce the human pancreas' natural insulin control mechanisms. The research team developed a Markov Decision Process (MDP) model which represents blood glucose levels as system states and uses insulin infusion as control actions to discover optimal insulin delivery through repeated interactions with a simulated human body model. The study develops reward functions through bioinspired methods which include two separate systems that combine long-term and short-term components to simulate pancreatic β-cell insulin production during fasting and after meals. The model incorporates drug response characteristics through specialized discount factors which enable the system to model both the delayed appearance of insulin effects and the total drug impact that accumulates during time. The Proximal Policy Optimization (PPO) algorithm trains the reinforcement learning policy through value networks which estimate both immediate and future results to create stable decisions that adapt well to unexpected situations.

The experimental evaluation used the FDA-approved UVA/Padova simulator to prove that the proposed system operates complete automated glucose control without users needing to announce their meals. The controller maintains blood glucose levels within the target range during both fasting and postprandial phases while it handles difficult situations which include sensor noise and insulin sensitivity changes and dawn phenomenon. The system maintains a high Time-in-Range (TIR) performance of approximately 89 to 92 percent while it reduces the occurrence of hypoglycemic events which shows its strong operational stability and trustworthiness. The model achieves intelligent glucose level regulation through its interpretability analysis which uses layer-wise relevance propagation to combine glucose measurements with rate of change information and insulin-on-board (IOB) data for safe dosing decisions without stacking insulin. The system performs well when it needs to handle different patient situations because it maintains consistent results through regular physiological changes and outside interference.

2.2 Summary of Literature Study:

Table. 2.1 literature review

SNo	Title& Publication Details	Agents/ Technologies Used	Work done	Remarks
1	Integrating Neural Differential Forecasting with Safe Reinforcement Learning for Blood Glucose Regulation	NeuralODE, Safe Reinforcement Learning, Probabilistic Safety Layer	Developed safe RL framework for insulin dosing with glucose forecasting.	Validated only in simulations.
2	An Improved Strategy for Blood Glucose Control Using Multi-Step Deep Reinforcement Learning	Multi-step DQN, Prioritized Replay	Proposed multi-step DRL model for delayed insulin effect handling. Improved glucose stability.	Real-life lifestyle factors were not considered
3	Fully-Automated PatientAgnostic Diabetes Management with Deep Reinforcement Learning	Soft Actor-Critic (SAC), Deep RL	Built patient-agnostic automated insulin delivery model. Achieved 80.3% TIR.	Patient diversity in evaluation remains limited.
4	Reinforcement Learning: A Paradigm Shift in Personalized Blood Glucose Management for Diabetes (2024)	PPO, Reinforcement Learning	Designed PPO-based closed-loop glucose control system. Achieved 73% TIR.	Tested on only 10 patients.
5	A safe-enhanced fully closed-loop artificial pancreas controller based on deep reinforcement learning	PPO, Deep Reinforcement Learning methods	Developed a DRL-based closed-loop controller	overriding unsafe policy, Incorporates a safety layer reducing hypoglycemia risk during aggressive insulin dosing.

6	Privacy-Preserving Glycemic Management in Type 1 Diabetes: Development and Validation of a Multiobjective Federated Reinforcement Learning Framework	FRL Methods, SAC	Developed a multi-step Deep Q-Network	Struggle to generalize across diverse patient profiles due to dependence on specific training conditions.
7	Deep Reinforcement Learning Control of Type-1 Diabetes with Cross-Patient Generalization	DDPG, RL controller	Proposed a DDPG-based RL controller	Less exploration in highly dynamic conditions.
8	A Hybrid Closed-Loop Blood Glucose Control Algorithm with a Safety Limiter Based on Deep Reinforcement Learning and Model Predictive Control	MPC, DRL framework	Designed a hybrid DRL-MPC framework	Multiple modules increases system complexity
9	Residual Hybrid Deep Learning Approach for Personalized Insulin Dose Estimation	LSTM,GRU	Developed a residual model . Achieved r^2 score 0.805	Clinical validation missing, Lack of dataset transparency
10	Toward a Fully Automated Artificial Pancreas System Using a Bioinspired Reinforcement Learning Design: In Silico Validation	Bioinspired RL,MDP,PPO	Developed a bioinspired reinforcement learning–based artificial pancreas system	Requires careful tuning of PK/PD parameters , High training complexity due to multiple reward functions

CHAPTER-3

PROPOSED METHODOLOGY

3.1 DESIGN METHODOLOGY

The proposed system uses reinforcement learning as its foundation to develop an automated insulin delivery system which operates within a simulated body system. The system operates through a simulated environment which mimics actual glucose-insulin interaction patterns. The learning agent receives patient information through four basic state variables which include blood glucose levels and insulin-on-board (IOB) values and body weight and rate of change. The system receives normalized states which the ensemble reinforcement learning framework uses to produce the best insulin dosing recommendations.

The design methodology focuses on three core components: the state space definition, the action space constraints, and the reward function design.

3.1.1 State Space (S(t))

The state vector contains physiological information together with time-based data which allows the agent to make secure insulin dosage selections without needing patients to report their food intake. The state at time step consists of four normalized features:

$$s_t = [g_t, \Delta g_t, \mathrm{IOB}_t, bw] \qquad \text{-(Eq 1)}$$

1. **Current Blood Glucose (BG(t)):** The Continuous Glucose Monitor (CGM) delivers real-time glucose readings which show the patient's present blood sugar level.

2. **Rate of Change (ΔBG(t)):** The glucose levels experience changes throughout the day as shown by recent data which tracks blood sugar elevation patterns and helps identify when food consumption causes blood sugar spikes.

3. **Insulin On Board (IOB(t)):** The pharmacokinetic model produces active insulin predictions which help users avoid dangerous insulin buildup that leads to hypoglycemia.

4. **Body Weight (BW):** The model contains a patient-specific physiological element which determines how insulin affects the body to create different dosage requirements for each person.

3.4.1 Action Space (A(t))

The action space in this project is continuous and represents the insulin dose administered by the insulin pump at each time step. The agent generates actions based on the current physiological state without requiring explicit meal announcements, enabling fully automated control:

Raw Action: The ensemble agent outputs a continuous value $a_{\text{raw}} \in [-1,1]$, representing a normalized control signal.

Action Transformation: This raw action is linearly scaled to a physiological insulin dose using cohort-specific clinical limits:

$$\text{Insulin} = \left(\frac{a_{\text{raw}}+1}{2}\right) \times I_{max} \quad \text{-(Eq 2)}$$

Where $I_{\text{max}}$is the maximum allowable insulin dose, defined based on patient cohort (child, adolescent, adult).

Nonlinear Scaling (for Safety): For child and adolescent cohorts, a squaring function is applied to the normalized action to prioritize smaller, safer doses while still allowing larger corrections when necessary:

$$\text{Insulin} = \left(\frac{a_{\text{raw}}+1}{2}\right)^2 \times I_{max} \quad \text{-(Eq 3)}$$

Safety Integration: The computed insulin dose is further processed through a safety layer that enforces clinical constraints such as hypoglycemia prevention, predictive glucose control, and insulin-on-board limits before being applied to the environment.

3.1.3 Reward Function (R(t))

The reinforcement learning agent depends heavily on its reward function design because this function needs to protect patients from harm while maintaining their blood sugar levels within safe boundaries. The proposed system uses a multi-zone reward system to help the agent keep blood sugar levels between 70 and 180 mg/dL which doctors consider safe while giving extra weight to the ideal range of 80 to 140 mg/dL. The system consists of multiple parts which serve to achieve distinct control targets. The system delivers positive feedback when glucose levels stay within the target range but it activates strong hypoglycemia

penalties to protect patients from harm. The system enforces growing penalties for high blood sugar levels which increase based on the degree of blood sugar deviation from the target range.

At each timestep t, the reward is defined as:

$$r_t = r_{\text{zone}} + r_{\text{hypo}} + r_{\text{hyper}} + r_{\text{trend}} + r_{\text{stability}} + r_{\text{IOB}} \quad \text{-(Eq 4)}$$

where each component represents a specific control objective. The term r_{zone}provides positive reinforcement when glucose remains within the target range, while r_{hypo}imposes severe penalties for hypoglycemia to ensure patient safety. The r_{hyper}term applies progressive penalties for hyperglycemia based on deviation from the desired range.

The system guides glucose levels to reach their target value through its design which also delivers rewards when blood sugar levels stay stable without sudden changes. The component operates insulin-on-board control to stop insulin buildup which produces future hypoglycemia risks. The system gives out extra rewards to users who achieve steady control and successfully manage their blood sugar levels when they become unstable. The system applies reward value limits through pre-set boundaries which help the learning process achieve both stability and fast learning results.

Positive Reward: The system gives users high rewards when their blood sugar levels stay between 80 and 140 mg/dL and users get extra bonuses when they keep their blood sugar levels close to the target value.

Negative Penalty: The medical system enforces strict penalties against patients who develop hypoglycemia at levels below 70 mg/dL with doctors focusing on punishing people who reach dangerous blood sugar numbers under 54 mg/dL while doctors apply increasing penalties to patients who develop hyperglycemia at levels above 180 mg/dL. The treatment system applies extra penalties when patients experience quick blood sugar changes and when their insulin levels exceed recommended values and their glucose levels move past the acceptable target range which helps doctors maintain patient.

3.1.4 Safety-Constrained Control Mechanism :

The system needs a safety-based control layer which runs as its foundation to deliver insulin safely to patients in clinical settings. The system applies action modification to block

dangerous insulin decisions by stopping all agent activities before they become operational.

The safety mechanism enforces critical physiological constraints. First, insulin delivery is completely suspended when blood glucose falls below a predefined hypoglycemic threshold, preventing dangerous low glucose conditions. In addition, a predictive safety check is performed using:

$$g_{t+\Delta} = g_t + \Delta g_t \cdot \Delta t \qquad \text{-(Eq 5)}$$

where Δt represents a short prediction horizon. If the predicted glucose level falls below a critical threshold, insulin delivery is halted to proactively prevent hypoglycemia.

Furthermore, insulin-on-board (IOB) constraints are applied to avoid insulin stacking:

$$a_t = \begin{cases} 0, if\ IOB_t \geq IOB_{max} \\ a_t, otherwise \end{cases} \qquad \text{-(Eq 6)}$$

The system decreases insulin delivery during fast glucose level decreases to stop blood sugar from falling any more. The safety constraints need to match patient groups because their insulin sensitivity differs between children and teenagers and adults. The system maintains equilibrium between its adaptive control system and patient safety requirements through these operational mechanisms which enable it to function as a dependable closed-loop insulin delivery system.

3.2 System Architecture Diagram

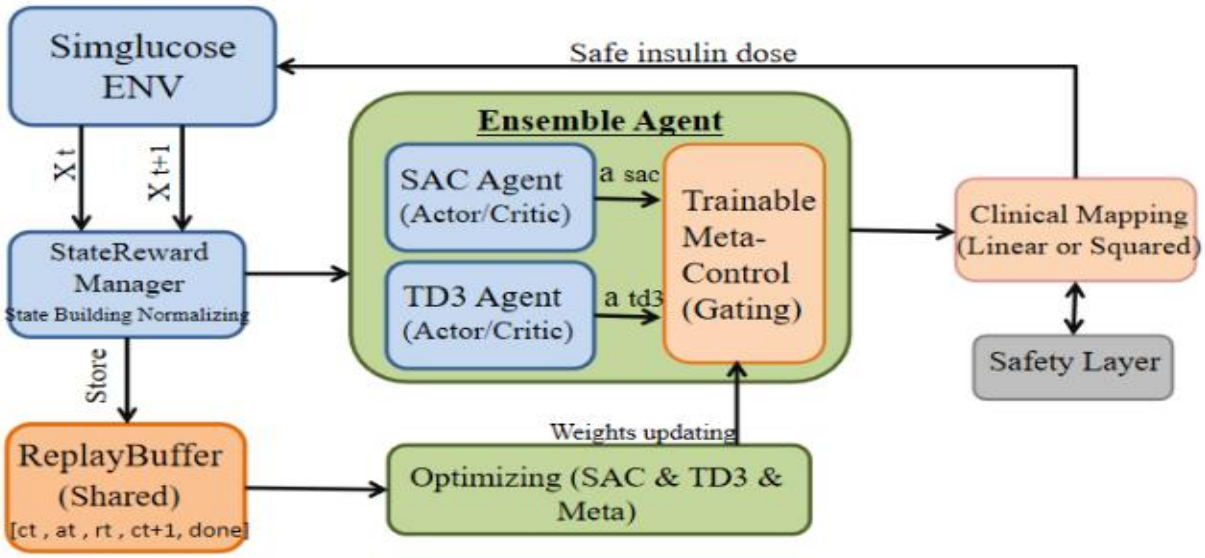

Figure 3.1 Architecture Diagram

The closed-loop artificial pancreas system architecture operates through an ensemble framework which uses reinforcement learning to deliver automatic insulin treatment inside the Simglucose simulation platform. The system processes all original physiological data which includes Continuous Glucose Monitor (CGM) data and insulin administration records and individual patient data about their body weight through the state formulation module. The module calculates glucose change rates (ΔBG) and produces Insulin-on-Board (IOB) values through its pharmacokinetic model which creates a structured state vector from normalized features.

Following state formulation, two actor-critic agents operate together as an ensemble through Soft Actor-Critic (SAC) and Twin Delayed Deep Deterministic Policy Gradient (TD3) to produce insulin dosing decisions. The two agents work together to generate separate action decisions which they derive from their present state information. The system uses a trainable meta-controller to create optimal control signals through weight-based action combination. The system activates its adaptive weighting system which distributes SAC exploration power to TD3 controller stability for improved decision-making performance under various physiological conditions.

The safety layer applies clinical restrictions to the combined action which prevents hypoglycemia and maintains predictive glucose control and restricts insulin-on-board amounts for safe insulin delivery. The system learns from a shared experience database which keeps track of events that both base agents and the meta-controller use for their training process. The training system receives realistic meal simulations with fixed patterns which researchers use to test their models against actual human body variations.

The system achieves glucose control through its adaptive closed-loop system which uses ensemble reinforcement learning together with safety-based control systems and physiological models to prevent both hypoglycemia and hyperglycemia.

3.3 Description of Algorithms

The proposed closed-loop insulin delivery system employs Deep Reinforcement Learning (DRL) algorithms to handle the complicated non-linear glucose-insulin relationships which occur in patients who have Type 1 Diabetes. The system achieves blood glucose control through its ability to modify insulin delivery amounts by following real-time physiological data which includes Blood Glucose levels (BG(t)), Rate of Change (ΔBG(t)), Insulin-on-Board (IOB(t)), and Body Weight (BW). The framework core consists of three essential components which include Soft Actor-Critic (SAC) and Twin Delayed Deep Deterministic Policy Gradient (TD3) and a Trainable Ensemble

Agent with a Meta-Controller. The previous research work used a Residual Hybrid LSTM-GRU model which operated on the AIM94 dataset to predict insulin doses. The model succeeded in tracking temporal relationships between glucose and insulin values through its recurrent architectural design. The system depended on supervised prediction but it failed to handle situations which demanded instant decision-making. The present DRL-based ensemble system learns its optimal insulin control policies through environmental interaction which produces a closed-loop system that enhances both system robustness and safety performance and clinical applicability.

1. **Residual Hybird Model :**

   The Residual Hybrid Network is a time-series deep learning model designed to predict insulin dosage by capturing both short-term and long-term temporal dependencies in physiological data. It combines Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures with residual connections to enhance learning stability and prevent information loss. This hybrid design enables the model to effectively process glucose trends, insulin history, and temporal features, allowing it to learn complex glucose-insulin dynamics. In this project, the model generates accurate and stable insulin dose predictions by leveraging sequential patterns in patient data. Its use of residual learning and recurrent structures makes it robust and well-suited for handling dynamic and nonlinear physiological variations.

Algorithm 1 Residual Hybrid Network for Insulin Dose Prediction

Require: Patient time-series dataset D
Ensure: Predicted regular insulin dose $\hat{Y}$

1: **Load and preprocess** D
   a) Extract glucose, insulin, and event features
   b) Encode temporal features (Hour, Minute, DayOfWeek)
   c) Generate lag, rolling, and interaction features
   d) Normalize features
2: Create sliding windows of length W for each patient
3: Split data into training, validation, and test sets
4: **Define model architecture**
   a) LSTM block ($64 \rightarrow 32$) with residual projection and addition
   b) GRU block ($32 \rightarrow 16$) with residual on last timestep
   c) Apply Layer Normalization and Dropout after each recurrent block
   d) Dense layers ($32 \rightarrow 16 \rightarrow 1$) for final regression output
5: Compile model (Adam optimizer $\eta = 10^{-3}$, MSE loss); monitor MAE
6: Train with early stopping and cosine-annealed learning rate
7: **return** predicted insulin dose $\hat{Y}$ for unseen data

Algorithm 1 – Residual hybrid network LSTM-GRU

2. Ensemble Model (SAC-TD3):

The Ensemble SAC-TD3 Agent is a hybrid reinforcement learning framework designed to combine the strengths of both stochastic and deterministic policies for optimal insulin control. It integrates the exploratory capabilities of the Soft Actor-Critic (SAC) agent with the stable and precise control of the TD3 agent through a dynamic weighting mechanism. At each timestep, both agents generate candidate actions based on the current physiological state, which are then combined using learned weights to produce a final ensemble action. This action is passed through a safety layer to ensure clinically safe insulin delivery before being applied to the environment. The system is trained using a shared replay buffer, allowing both agents to learn from common experiences while improving coordination. This ensemble approach enhances robustness, balances exploration and stability, and results in more reliable and adaptive glucose regulation across varying conditions.

Algorithm 2 Ensemble SAC-TD3 Agent Training and Evaluation

Notation: $\pi_\phi^{SAC}, \pi_\theta^{TD3}$: Actor; $Q_{\psi_{1,2}}^{SAC}, Q_{\omega_{1,2}}^{TD3}$: Critic; $\mathcal{D}$: Buffer; M: Manager; S: Safety Layer; $s_t = [\mathrm{BG}_t, \Delta\mathrm{BG}_t, \mathrm{IOB}_t]$; C_t^{SAC}, C_t^{TD3}: Confidence scores

Phase 1: Initialization

for each patient $p \in \mathcal{P}$ **do**
  Initialize environment, SAC/TD3 agents, shared buffer $\mathcal{D}_p$, Manager M_p, Safety Layer S_p
  Set initial ensemble weights $w_{SAC} = 0.6, w_{TD3} = 0.4$
end for

Phase 2: Training

for each patient $p \in \mathcal{P}$ **do**
  for episode = 1 to max_episodes **do**
    for $t = 1$ to max_timesteps **do**
      Construct state s_t using M_p
      Sample $a_t^{SAC} \sim \pi_{\phi_p}^{SAC}(\cdot|s_t)$; $a_t^{TD3} = \pi_{\theta_p}^{TD3}(s_t) + \epsilon$
      Compute confidence scores C_t^{SAC}, C_t^{TD3}; fuse actions: $a_t^{ensemble} = w_t^{SAC} a_t^{SAC} + w_t^{TD3} a_t^{TD3}$
      Apply Safety Layer: $a_t^{safe} = S_p(a_t^{ensemble}, s_t)$; execute, observe r_t, s_{t+1}
      Store $(s_t, a_t^{ensemble}, r_t, s_{t+1}, \text{done})$ in $\mathcal{D}_p$
      Update SAC and TD3 networks by sampling from $\mathcal{D}_p$
    end for
  end for
  Save ensemble model $\{\phi_p, \theta_p, \psi_{1,2}, \omega_{1,2}\}$
end for

Phase 3: Evaluation

for each patient $p \in \mathcal{P}$ **do**
  Load trained actors $\{\phi_p, \theta_p\}$; evaluate on fixed scenario to compute performance metrics (e.g., TIR%).
end for

Algorithm 2 -Ensemble model SAC-TD3 Agent

3. SAC AGENT :

The Soft Actor-Critic (SAC) model operates as a stochastic off-policy reinforcement

learning system which works to achieve maximum expected rewards together with maximum policy entropy. The system achieves successful exploration of continuous action spaces which proves essential for handling different body states that appear during meal interruptions and blood sugar level changes. The project uses SAC to produce smooth exploratory insulin dose actions which help the system find different control methods that work well. The system learns through entropy-based methods which make it suitable for handling unpredictable situations that change rapidly.

Algorithm 1 SAC Agent Training and Evaluation

Notation: π_ϕ: Actor; Q_ψ: Critics; D: Buffer; S: Safety Layer; M: State Manager

Phase 1: Initialization

1: $\forall p \in \mathcal{P}$: Initialize environment, π_ϕ, Q_ψ, D_p; Copy targets $\bar{\psi} \leftarrow \psi$.

Phase 2: Training

2: for each patient $p \in \mathcal{P}$, episode $e = 1 \ldots E$ do
3: for $t = 1 \ldots T$ do
4: $s_t \leftarrow M(o_t)$; Sample $a_t \sim \pi_\phi(\cdot|s_t)$ (or random if warm-up)
5: Apply Safety: $a_t^{safe} \leftarrow S(a_t)$; Step env: $s_{t+1}, r_t \leftarrow Env(a_t^{safe})$
6: Store transition (s_t, a_t, r_t, s_{t+1}) in D_p
7: Update Q (MSE loss) and π (maximize $Q - \alpha \log \pi$)
8: Soft update target networks: $\bar{\psi} \leftarrow \tau\psi + (1-\tau)\bar{\psi}$
9: end for
10: end for

Phase 3: Evaluation

11: $\forall p$: Load ϕ_p, run fixed scenario, report TIR, Hypo%, Hyper%.

Algorithm 3 – Soft Actor Critic agent

4. **TD3 AGENT :**

The Twin Delayed Deep Deterministic Policy Gradient (TD3) model functions as a deterministic off-policy reinforcement learning system which generates accurate control responses for continuous action environments. The system solves actor-critic method overestimation problems through clipped double Q-learning which operates by using two critic networks to determine target values through their minimum estimate. The training process of TD3 gets better stability because the method delays actor network updates which happen at different times than the critic network updates. The system uses this

approach to cut down on policy learning variance which results in more dependable policy creation.

Algorithm 2 TD3 for Stable Insulin Control

Input: State vector $s_t = [\mathrm{BG}_t, \Delta\mathrm{BG}_t, \mathrm{IOB}_t]$
Output: Deterministic action a_t
Initialize Actor μ_θ and Twin Critics $Q_{\omega_{1,2}}$
Input State s_t into Actor network
Forward Pass:
    $a_{raw} = \mu_\theta(s_t)$
Action Processing:
if Training **then**
    $a_t = \mathrm{clip}(a_{raw} + \epsilon, -1, 1)$, where $\epsilon \sim \mathcal{N}(0, \sigma)$
else
    $a_t = a_{raw}$
end if
Training Step (Delayed):
    Update Q_ω via Bellman error (using min target Q)
    If $step \mod d == 0$: Update θ to maximize $Q_{\omega_1}(s, \mu_\theta(s))$
return a_t

Algorithm 4 – Twin Delayed Deep Deterministic agent

5. Meta Ensemble Agent :

The Ensemble Agent combines the outputs of SAC and TD3 using a trainable meta-controller that dynamically assigns weights based on the current physiological state. Instead of relying on fixed or heuristic combinations, the meta-controller learns to balance exploration (SAC) and stability (TD3) in a state-dependent manner. The combined action is then passed through a safety layer to enforce clinical constraints before being applied to the environment.

The meta-controller is implemented as a lightweight neural network that takes the normalized state as input and outputs adaptive weighting coefficients for both agents, ensuring that their contributions sum to one. This allows the system to emphasize exploratory actions during uncertain or rapidly changing conditions (such as post-meal spikes) while favoring stable and conservative actions during steady-state periods. Both SAC and TD3 agents are trained using a shared replay buffer, enabling efficient learning from common experiences and improving coordination between the policies.

During training, the meta-controller is optimized using feedback from the critic network,

allowing it to learn which agent performs better under specific physiological conditions. This results in a context-aware decision-making process that adapts over time. By leveraging the complementary strengths of stochastic and deterministic policies, the ensemble approach enhances robustness, reduces glucose variability, and improves overall safety. This architecture ultimately provides a more reliable and adaptive closed-loop insulin control system compared to individual agents.

Algorithm 1 Meta-Ensemble Policy Optimization (MEPO)

Notation: π_{sac}, π_{td3}: Base Agents; $\mathcal{M}_\theta$: Meta-Controller; $\mathcal{S}$: Safety Layer; $\mathcal{D}$: Buffer; $w_{s,t}$: Weights.

Phase 1: Initialization

Initialize: $\{\pi_{sac}, \pi_{td3}\}$ and $\mathcal{M}_\theta$ with weights θ; empty Buffer $\mathcal{D}$; cohort $\mathcal{P}_{adult,adolescent,child}$.

Phase 2: Training

for episode = 1 to max_episodes **do**
  Sample patient $p \sim \mathcal{P}_{cohort}$ and initialize environment Env_p.
  Reset Env_p, obtain initial physiological state s_0 and normalize to c_0.
  for $t = 1$ to max_timesteps **do**
    Compute weights $[w_{sac}, w_{td3}] \leftarrow \mathcal{M}_\theta(c_t)$ with 20% safety clamping.
    Generate ensemble action: $a_t = w_{sac}\pi_{sac}(c_t) + w_{td3}\pi_{td3}(c_t)$.
    Apply Clinical Mapping and Safety Layer: $\hat{a}_t \leftarrow \mathcal{S}(a_t, s_t)$.
    Execute $\hat{a}_t$, observe s_{t+1}, and calculate metabolic reward r_t.
    Store transition $(c_t, a_t, r_t, c_{t+1}, \text{done})$ in $\mathcal{D}$.
    if $|\mathcal{D}| >$ learning_starts **then**
      Sample mini-batch from $\mathcal{D}$ and update base agents $\{\pi_{sac}, \pi_{td3}\}$.
      Update Meta-Controller $\mathcal{M}_\theta$ by $\nabla_\theta \mathbb{E}[\min Q_{sac}(c_t, a_{ens})]$ via backprop.
      Save *best_model* if rolling cohort reward improves.
    end if
    if done **then break**
    end if
  end for
end for

Phase 3: Evaluation

Evaluate: Test $\mathcal{M}_\theta$ ensemble on cohort $\mathcal{P}_{test}$; report TIR, Hypo%, and Hyper% clinical metrics.

Algorithm 5 – Meta Ensemble Policy Optimization

3.4 Simulation Environment and Requirement Specification

3.4.1 Simulation Environment (Data Source)

Instead The system operates through a simulation platform which provides exact simulations to replace its dependency on static data sets for training and evaluation purposes. The data gets produced through real-time generation by SimGlucose[13] which functions as a Python-based version of the FDA-approved UVa/Padova Type 1 Diabetes Simulator. The simulator delivers an authentic glucose–insulin interaction model which has received clinical validation to support reinforcement learning-based insulin control system

development through safe methods.

The simulation environment models essential physiological processes such as glucose absorption, insulin kinetics, and carbohydrate digestion. The system enables virtual patient management through multiple patient groups which contain adult and adolescent and child patient categories. The system learns to create control policies which adapt to different patient needs by using individual physiological data points from each patient including their body mass and their ability to react to insulin. The training process uses various patient data which helps the model build strong performance abilities without learning particular patient details.

The training process includes a stochastic meal scenario which helps the model improve its ability to generalize. The system operates with random meal scheduling and unpredictable carbohydrate amounts which create various blood sugar challenges for the agent to face. The system enables the agent to develop flexible methods for dealing with actual environmental unpredictability. For evaluation, all models operate under the same testing environment which enables performance comparison between different models. The simulation environment operates as a dependable and adaptable data source which allows reinforcement learning agents to stay active while they develop their abilities and receive performance evaluations.

3.4.2 Meal Generation Scenario

To simulate the realistic behavior of glucose-insulin dynamics, a stochastic approach for generating meal scenarios[4] is integrated into the simulation process. Meal generation plays an important role in analyzing the effectiveness of the reinforcement learning model since glucose levels are influenced significantly by meal consumption. Meal generation is done using a probabilistic model based on that used by Wang et al[12]. In this case, several meals are scheduled randomly during the day, where each meal has a different time of occurrence and carbohydrate content. In particular, there are six possible meal scenarios over a period of 24 hours, including breakfast, snacks, lunch, and dinner. Meals are simulated based on their respective probability distribution:

The probability of meal occurrence is defined as:

- $p = [0.95, 0.3, 0.95, 0.3, 0.95, 0.3]$,

In other words, high probabilities indicate large meals, and low probabilities correspond to possible snack events.

The possible times between meals have physiological plausibility limits. Specifically, the time of occurrence of each meal k is chosen from a truncated normal distribution, which is defined by mean time μ_t, variance σ_t, and boundaries lower(lo) and upper(up). In this way, the meals' timings will reflect a physiologically plausible scenario, yet randomness will still be maintained. The parameters used for the definition of the meals' time occurrence are as follows:

- Lower bounds: $lo = [5, 9, 10, 14, 16, 20] \times 60$
- Upper bounds: $up = [9, 10, 14, 16, 20, 23] \times 60$
- Mean times: $\mu_t = [7, 9.5, 12, 15, 18, 21.5] \times 60$
- Standard deviations: $\sigma_t = [60, 30, 60, 30, 60, 30]$

The number of carbohydrates per meal is determined by the body weight of each patient (BW). Thus, the physiologically plausible scenario can be achieved by defining a relation between BW and meal size. The amount of carbohydrates contained in a meal is chosen from a normal distribution:

- Mean meal size: $\mu_a = [0.7, 0.15, 1.1, 0.15, 1.25, 0.15] \times BW$
- Standard deviation: $\sigma_a = 0.15 \times \mu_a$

Thus, large meals imply higher carbohydrate numbers compared to snacks, which are comparatively smaller.

The meal generation process in general could be described as follows:

1. Initialize an empty meal set E.
2. For each meal slot k (k = 1 to 6):
    - Generate a random probability value p_{tmp}.
    - If $p_{tmp} \leq p[k]$, generate a meal event:

- Sample meal amount using a normal distribution.
- Sample meal time using a truncated normal distribution.
- Add the meal (amount, time) to the set E.

3. Return the final meal set E for the simulation episode.

Such an approach creates significant variation in terms of meal time and carbohydrate consumption, making the environment non-stationary. As a result, the meal disturbances will have to be inferred indirectly via glucose values observed by the agent.

The lack of meal announcement information implies that the learning algorithm must employ adaptive control only based on the physiological feedback. In this regard, the developed approach can be considered a fully closed-loop environment.

All together, the presented stochastic meal generation scenario adds an additional level of complexity to the problem by increasing its practicality.

3.4.3 Requirement Specification

To successfully reproduce the training and evaluation of the Ensemble SAC-TD3 agent, the following hardware and software specifications are required:

PyTorch: The project depends on PyTorch to function as its main deep learning framework. The system employs this tool to establish complex neural network structures which include the Actor networks with Mixture Density Network layers and the Critic networks. The Soft Actor-Critic entropy update requires gradient calculations which PyTorch's dynamic computation graph enables.

Gymnasium: Gymnasium provides the standard interface for Reinforcement Learning. The system operates through Simglucose which creates a simulation environment that shows glucose levels to RL agents and lets them choose insulin doses for each time step.

Simglucose: This is the core domain-specific library that simulates the patient's physiology. It solves the ordinary differential equations (ODEs) representing the glucose-insulin dynamics, providing the "Ground Truth" biological response to the agent's actions.

NumPy & Pandas: NumPy functions as a numerical processing tool which executes high-speed operations to handle replay buffer arrays and action transformation tasks. The software tracks training metrics through Pandas which also structures evaluation results

and produces patient performance data into CSV files.

Scikit-learn: The preprocessing stage uses Scikit-learn which provides tools for feature scaling and normalization and additional data analysis capabilities. The system enables fast numerical data processing while it generates appropriate inputs which reinforcement learning systems need to function.

Matplotlib: Matplotlib serves as the tool which displays simulation outcomes through its graphical presentation of blood glucose levels and insulin administration data and performance evaluation results. The tool allows users to present both model operations and evaluation results through distinct graphical displays.

TensorBoard: The training process of reinforcement learning agents becomes visible through TensorBoard which provides both monitoring and visualization functions. The tool shows how important metrics develop through time which includes reward values and loss statistics and how policies respond which helps with debugging and performance analysis.

Custom Safety Layer: A safety layer is implemented to enforce clinical constraints by limiting unsafe insulin actions, such as preventing hypoglycemia and excessive insulin dosing.

State Management Module: The state management system processes raw glucose data to compute features like rate of change and insulin-on-board (IOB), ensuring meaningful inputs for the learning agents.

Replay Buffer: The Replay Buffer keeps all previous experiences which include state information and action taken and received reward and following state to help reinforcement learning agents train effectively through random experience selection.

3.5 Hyperparameter Configuration and Training Strategy

The performance of the meta-ensemble reinforcement learning algorithm significantly depends on choosing appropriate hyperparameters for SAC and TD3 learners. This impacts the learning dynamics, convergence, and general stability of the resulting control policy. Appropriate tuning of the hyperparameters guarantees the ability of the agent to learn efficient policies for insulin dosing while ensuring safety and reliability of glucose management.

The choice of hyperparameters used in this research relies on common settings used in literature in regard to reinforcement learning. The main hyperparameters used for training purposes are listed in Table 3.1.

Table. 3.1 Hyperparameter for Proposed Meta-Ensemble RL Agent

Parameter	**Value**
Learning	3e-4
Discount Factor	0.99
Batch Size	256
Replay Buffer Size	1e6
Target Update Rate (τ)	0.005
Policy Noise (TD3)	0.2
Noise Clip (TD3)	0.5
Entropy Coefficient (SAC)	Automatic
Initial Ensemble Weights	SAC:0.6 , TD3:0.4

3.5.1 Training Strategy

Training proceeds via continuous interaction between the reinforcement learning algorithm and the simulation environment. In each time step, the algorithm perceives the current physiological state, takes an insulin action, and receives rewards based on the glucose level generated. Through such interactions, it progressively builds its skills for developing an optimal policy.

The experiences gathered in each interaction are saved in the experience replay buffer, where batches of them are randomly selected for learning. This approach promotes sample efficiency and makes learning more stable due to the lack of dependence among consecutive samples.

Training occurs under stochastic meals, meaning that there is randomness and variability in the environment. Furthermore, constraints are applied during training so that the agent does not perform actions that could result in hazardous outcomes. As such, the algorithms are expected to learn both efficient and safe policies.

3.5.2 Explanation of Key Hyperparameters

The following hyperparameters control different aspects of the learning process:

Learning Rate:

Controls how quickly the neural network will modify its weights during the training process. Lower learning rates provide smooth weight updates, which is crucial when using the system in real-world applications like medical control systems.

Discount Factor (γ):

Adjusts the weight between the short-term rewards and the long-term goals. A γ closer to 1 will emphasize long-term stability over short-term glucose levels.

Batch Size:

Defines how many experiences will be selected from the replay buffer during each training process. Large batch sizes will improve stability but require more computational resources.

Replay Buffer Size:

The size of the replay buffer can reduce correlation between experiences, promoting generalization by exposing the agent to a wide variety of situations.

Target Update Rate (τ):

Controls how quickly the target networks are updated. Lower target update frequencies lead to more stable training processes by minimizing sudden shifts in learning objectives.

Policy Noise and Noise Clip (TD3):

Allows some randomness when taking actions, which improves the exploration of the state-action space and prevents overfitting.

Entropy Coefficient (SAC):

It gets automatically adjusted during the training process, allowing a dynamic balance between exploration and exploitation.

Initial Ensemble Weights:

These are the weights the initial assigned to SAC and TD3 agents. Assigning slightly higher weight to SAC encourages exploration during the early stages of training, which is later refined by the meta-controller.

Both selected hyperparameters and training procedure ensure convergence and successful acquisition of policies. Combination of the described techniques like replay, randomness in environments, and constraints make the suggested approach able to provide consistent results while controlling glucose levels in a wide range of patients.

CHAPTER 4

RESULTS AND OBSERVATIONS

4.1 Deep Learning-Based Insulin Dose Prediction Using Residual LSTM-GRU

In order to set a benchmark to predict personalized insulin dosage using a Deep Learning approach, a residual sequence learning model consisting of LSTM-GRU architecture was built first. The dataset used for training the model included clinical data obtained from AIM94 dataset [14].

Performance of the model is estimated using conventional measures like R^2 Score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Mean Absolute Scaled Error (MASE). Results of the model are shown in Table 4.1.

Table-4.1 Performance Comparison of DL Models for Insulin Prediction (AIM94 Dataset)

Model	R2	MAE	MSE	MASE
LSTM – Transformer	0.53	1.70	10.26	0.60
GRU-Transformer	0.53	1.84	10.14	0.57
Baseline LSTM-GRU	0.62	1.43	8.29	0.32
Residual Sequence Learning Model	0.805	1.29	3.27	0.29

The residual sequence learning model had an R^2 Score of 0.805, which outperforms baseline models like LSTM-Transformer (0.53), GRU-Transformer (0.53), and the LSTM-GRU hybrid model (0.62). In addition to that, the MAE of the model was brought down to 1.29, and MSE was reduced to 3.27. To showcase how the performance of the model improved, comparisons for each of them are shown in Figures 4.1–4.3.

The results show that combining residual connections and enhanced feature representations considerably improves the network's ability to detect both short- and long-term variations in the glucose-insulin system. In addition, Figure 4.4 shows a comparison of real versus predicted values of insulin for a typical patient over a week-long period.

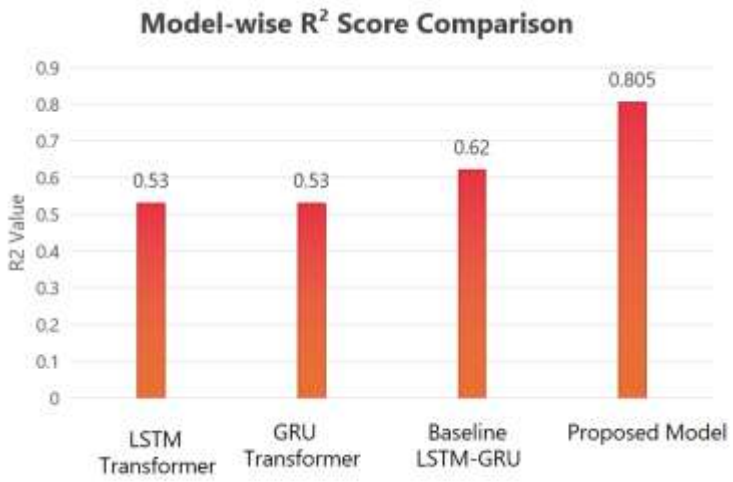

Figure 4.1. R² Score Comparison Across Deep Learning Models

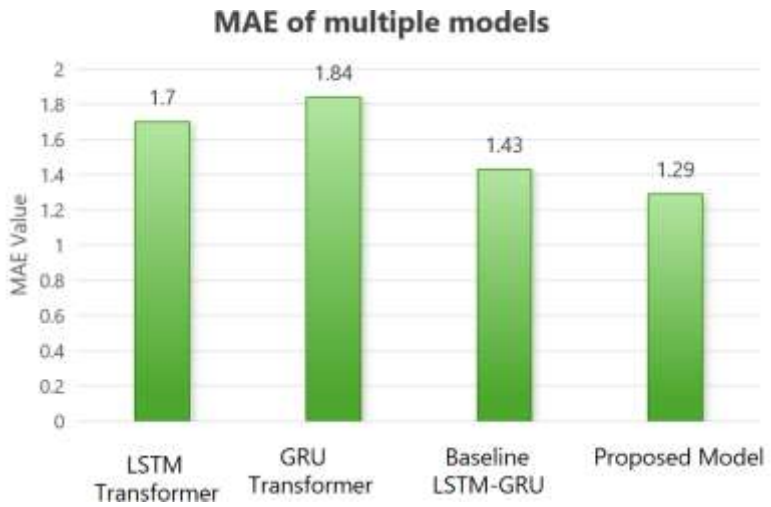

Figure 4.2. MAE Comparison Across Deep Learning Models

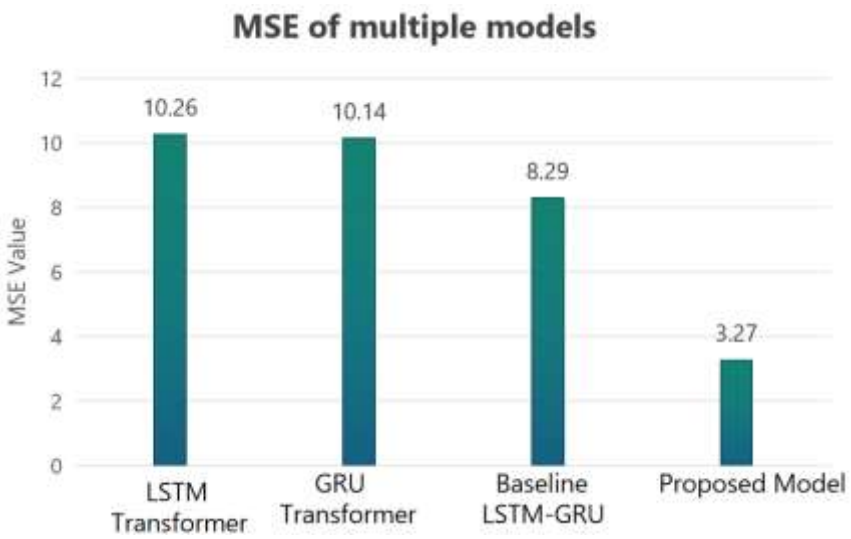

Figure 4.3. MSE Comparison Across Deep Learning Models

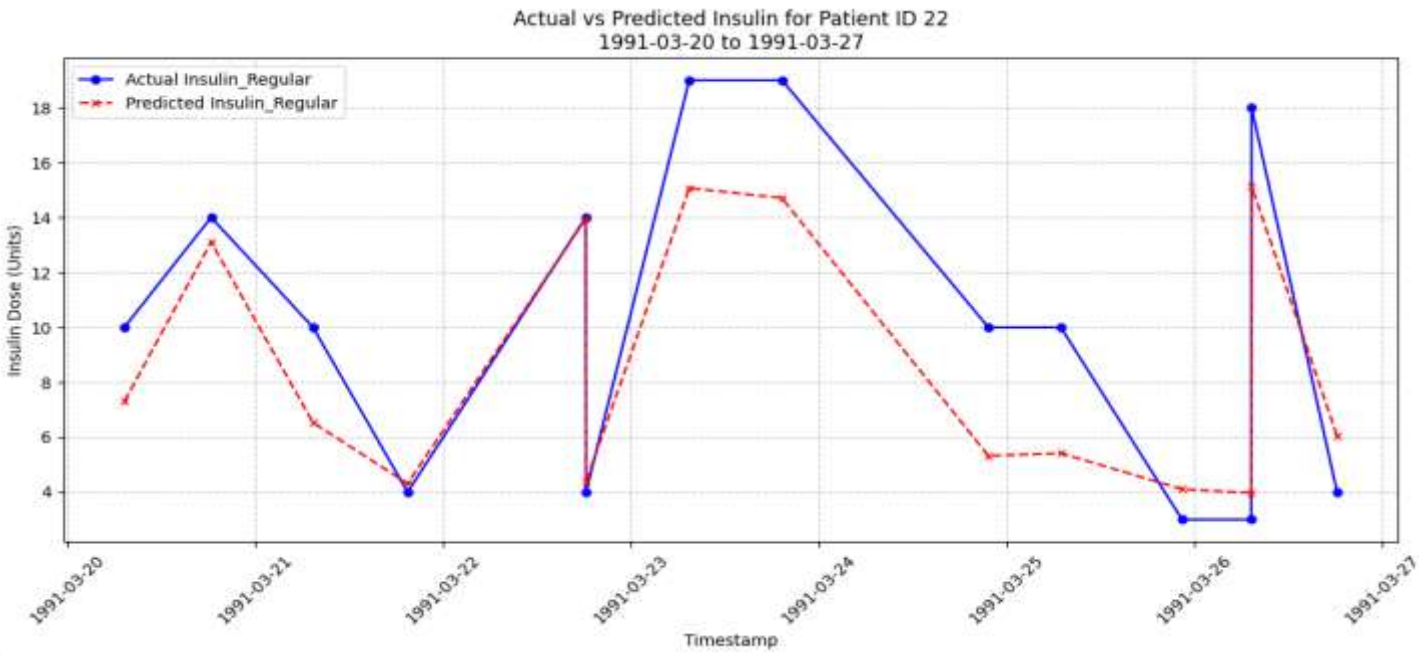

Figure 4.4. Actual vs Predicted Insulin Dose for a Patient

4.2 Initial Reinforcement Learning Ensemble Results

In order to further the predictive decision-making model, a reinforcement learning algorithm using Soft Actor-Critic (SAC) and Twin Delayed Deep Deterministic Policy Gradient (TD3) agents was initially developed. A weighted ensemble methodology was adopted in order to integrate the behaviors of both the SAC and TD3 agents to improve glucose regulation.

The performance comparison of the algorithm was performed in an adult population cohort where separate models were designed for each patient in isolation. A summary of the performance between various agents are shown in Table 4.2.

Table-4.2 Performance Comparison of RL Agents for Adult Cohort

Agents	**Mean Glucose(mg/dL)**	**Time In range(%)**	**Time in hypo(%)**	**Time in hyper(%)**
SAC-MDN	140.289	70.11	13.351	16.539
TD3	121.3677	72.26	22.09142	6.028316
Ensemble (SAC + TD3)	112.8184	78.78	11.8409	9.379509

The use of ensemble improves TIR to 78.78%, compared to SAC-MDN (70.11%) and TD3 (72.26%). Moreover, it manages hypoglycemia at 11.84%, against SAC-MDN (13.35%) and

TD3 (22.09%), while keeping hyperglycemia at a balanced level. The performance of different models in this experiment is shown in Figure 4.5.

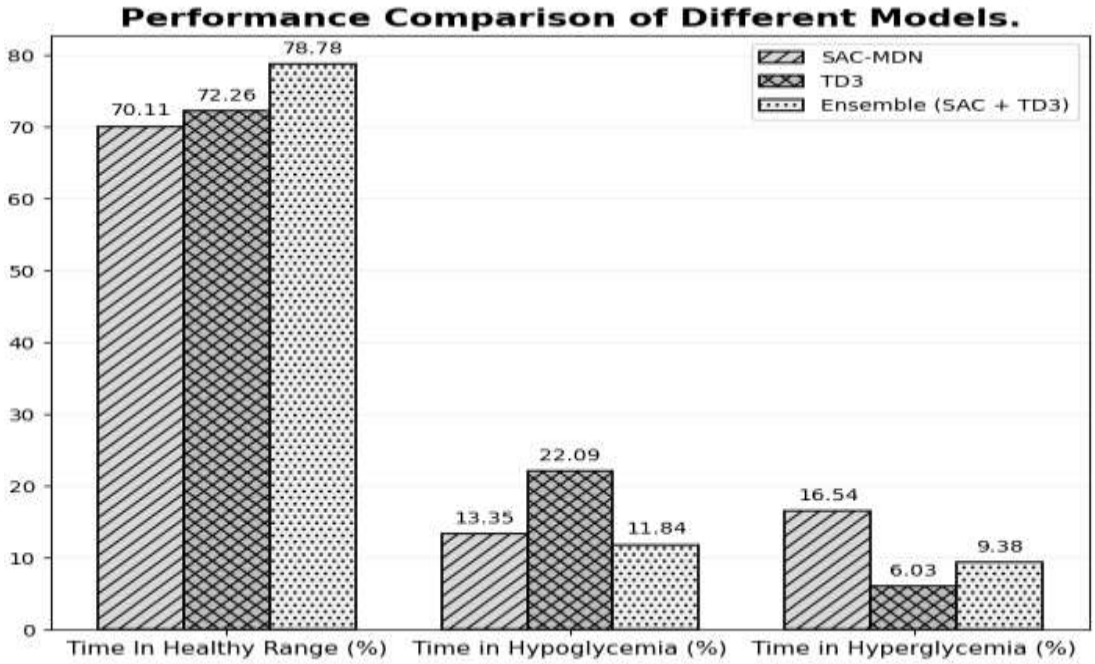

Figure 4.5. Comparison of TIR, Hypoglycemia, and Hyperglycemia Across RL Agents

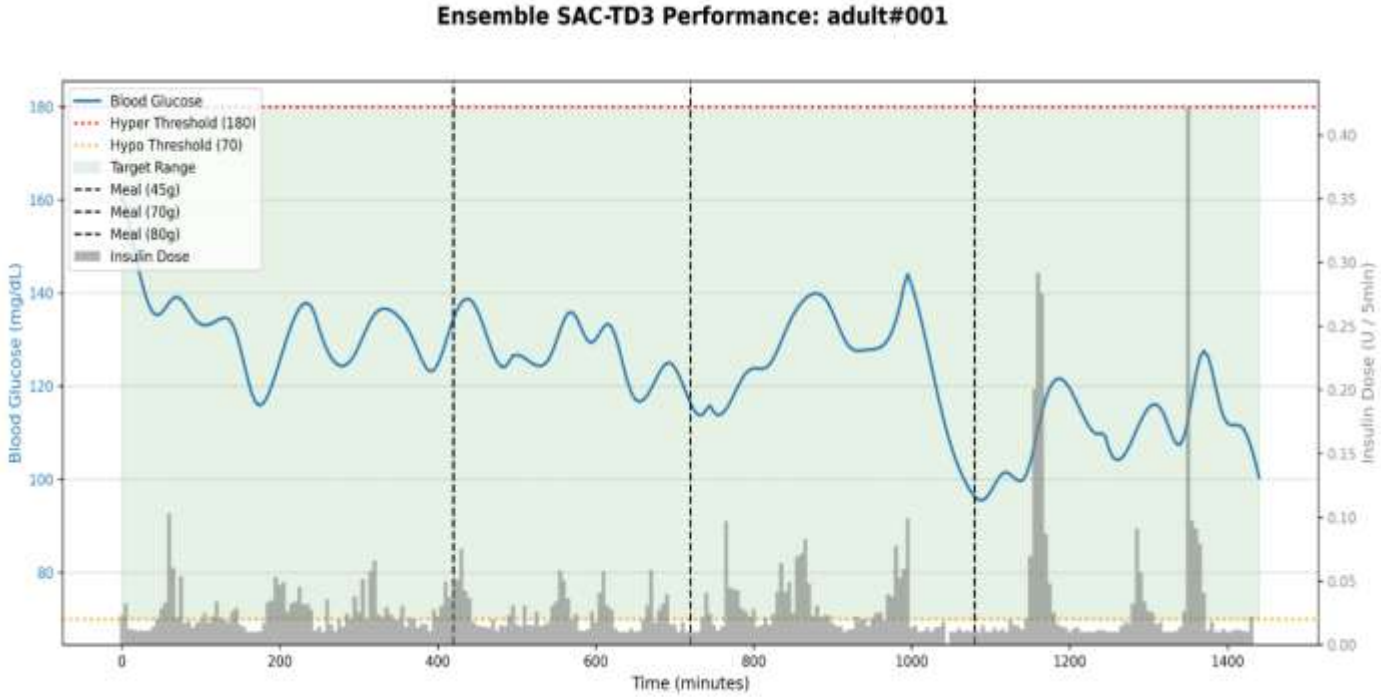

Figure 4.6. Glucose and Insulin Response of SAC-TD3 Ensemble Agent for an Adult Patient

The results show that using reinforcement learning in combination with the ensemble method improves glucose management than using agents separately. This methodology, however, is restricted to a single cohort (adult patients only). These findings have led us to develop our novel proposed methodology for a **cohort-aware** meta-ensemble approach.

4.3 Overall Performance Comparison

4.3.1. Evaluation Metrics

When evaluating the performance of models for reinforcement learning (RL) control applications, an appropriate measure will be the ability of the autonomous insulin therapy to keep BG within acceptable and clinically safe limits. The following performance measures obtained from closed loop simulations give us numerical measures about performance and reliability of the control policy:

Time In Range (TIR, %):

The most important clinical measure is the time in the target range between 70 mg/dL and 180 mg/dL for the patient's glucose concentration. It represents the key objective in the reward function of the RL agent.

Mean Glucose (mg/dL):

The average value of blood glucose of a patient over the time period. Closer the values to the optimal level (110 mg/dL), indicates better the control performance.

Time in Hypoglycemia (Time in Hypo, %):

It refers to the amount of time spent under hypoglycemia, when the glucose level falls under 70 mg/dL. Its minimization is the highest priority for any closed loop system.

Time in Hyperglycemia (Time in Hyper, %):

Time in hyperglycemia refers to the time spent when the glucose concentration is above 180 mg/dL.

4.3.2. Comparative Performance of Models

Evaluations of the suggested meta-ensemble reinforcement learning framework were carried out with respect to baseline SAC and TD3 against the realistic setting of meals in simulations. In Table 4.3, the mean performances across the three groups of adults, adolescents, and children are given.

Table-4.3 Performance Comparison Across Methods and Cohorts (short term-24hours)

Method	Group	TIR(%)	HYPO(%)	HYPER(%)
Meta-Ensemble Agent	Adult	68.72	5.30	25.99
	Adolescent	82.91	5.40	11.70
	Child	65.99	18.17	15.85
SAC Agent	Adult	64.88	5.23	29.17
	Adolescent	56.17	20.74	17.03
	Child	67.82	26.70	1.05
TD3 Agent	Adult	55.93	15.29	25.01
	Adolescent	70.39	7.80	14.49
	Child	64.87	25.17	5.89

According to **Table 4.3**, the ensemble-based approach gives better results than SAC and TD3 for most of the cohorts.

The adolescent group shows the best performance in terms of TIR achieved with an accuracy of 82.91%, which is significantly higher than SAC's (56.17%) and TD3's (70.39%). Meanwhile, the incidence rate of hypoglycemia is kept low at 5.4%, and the hyperglycemia is minimized (11.7%), implying effective glucose regulation.

The adult group exhibits a similar trend, with the TIR being 68.72% using ensemble, TD3 (55.93%), and SAC (64.88%). Similarly, the incidence rates of hypoglycemia and hyperglycemia are low, and hyperglycemia is less severe than SAC's results. However, the existence of moderate hyperglycemia implies the challenge of controlling delayed glucose dynamics among adults.

For the **child cohort**, TIR for the proposed ensemble equals 65.99%. Moreover, the use of the ensemble ensures a significant reduction in the rate of hypoglycemia in contrast to SAC (26.7%) and TD3 (25.17%). SAC provides lower hyperglycemia, but higher hypoglycemia levels.

Thus, it can be stated that the proposed ensemble shows better results concerning the trade-off between glycemic control and safety compared to the baseline therapies. In terms of performance stability, the adolescents provide the best results, while children show the highest variability due to the higher insulin sensitivity.

4.3.3. Visual Comparison of Performance Metrics

The comparative performances of the different models for different cohorts can be found from Figures 4.7 through 4.9.

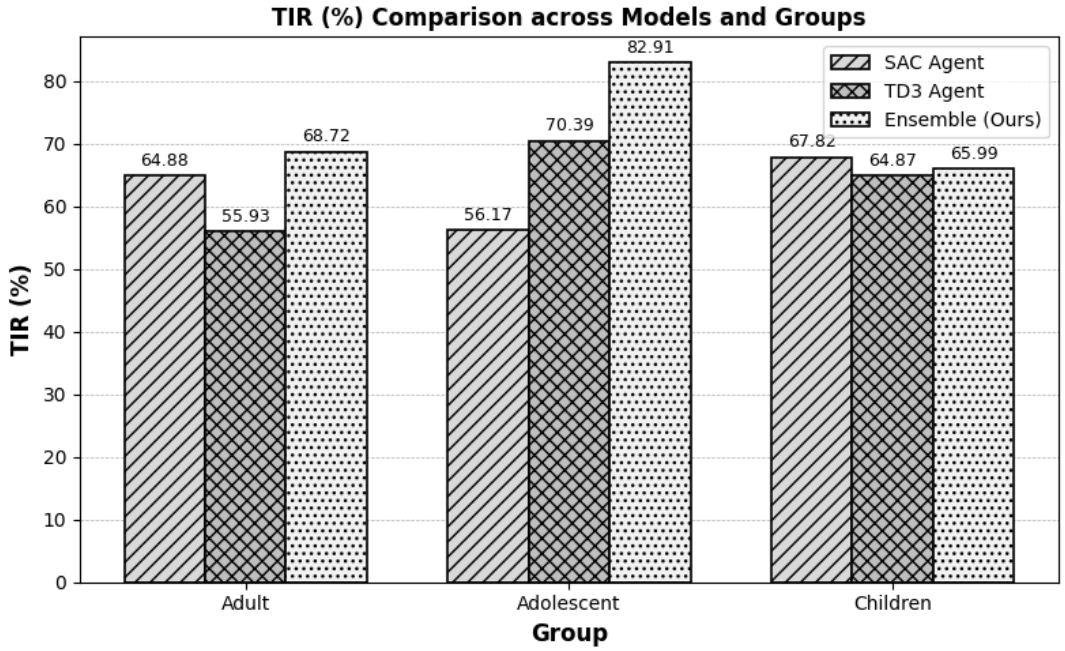

Figure 4.7. Comparison of Time in Range (TIR) Across Methods and Cohorts

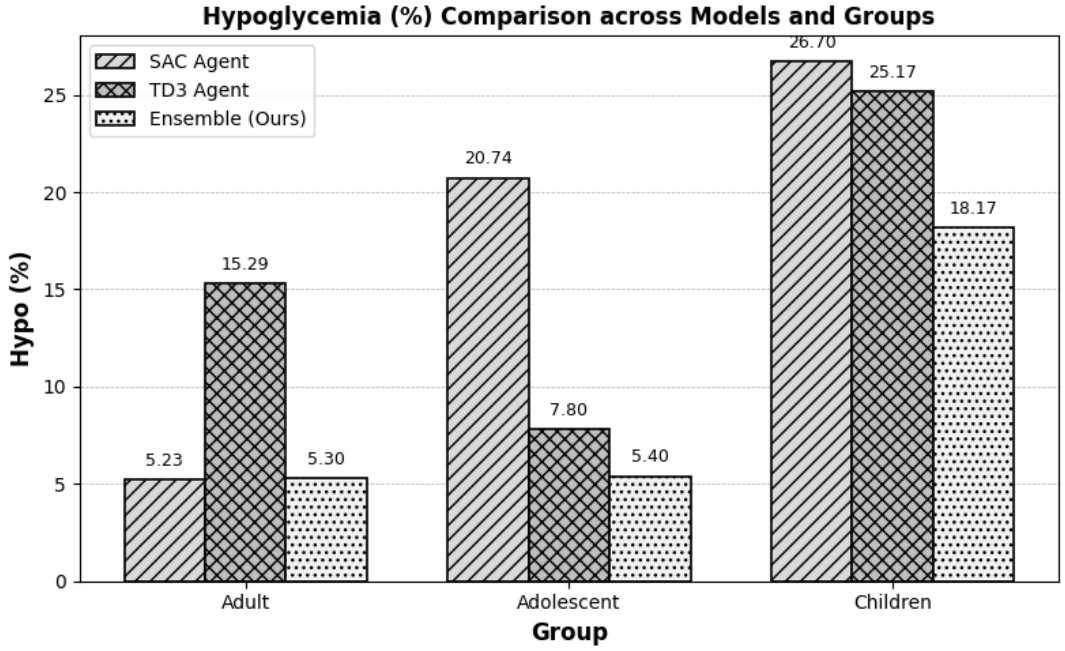

Figure 4.8. Comparison of Hypoglycemia Percentage Across Methods and Cohorts

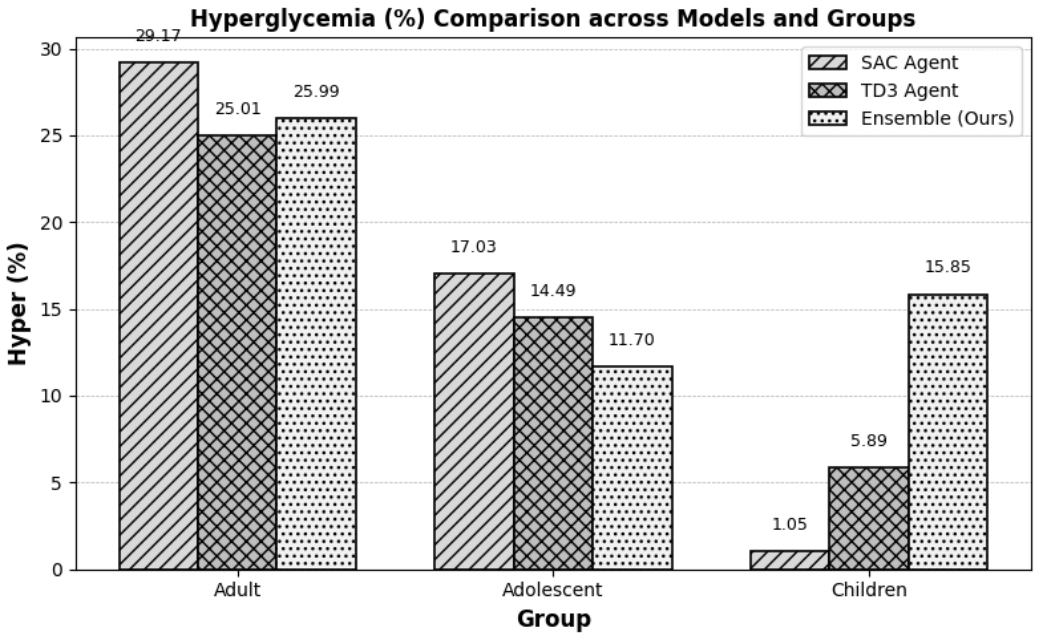

Figure 4.9 Comparison of Hyperglycemia Percentage Across Methods and Cohorts

The examination of time in range (TIR) reveals that the ensemble model obtains the highest amount of time in the desired range, and especially in the adolescent group. The analysis of the hypoglycemia reveals that the ensemble model is capable of reducing the hypoglycemic time significantly compared to baseline models, especially in the child group. Moreover, the hyperglycemia test also reveals that the ensemble model performs better at lowering the excessive blood glucose, specifically in the adolescent group.

4.4 Long-Term Performance Evaluation

4.4.1. Stability and Robustness Analysis

To analyze the robustness and stability of the developed model, we simulated it for a prolonged period of time equal to 10 days under realistic dietary inputs. The obtained results of the meta-ensemble method in different cohorts can be seen in Table 4.4.

Table-4.4 : Long-Term (10-Day) Performance of the Proposed Ensemble Model

Method	Group	TIR (%)	Hypo (%)	Hyper (%)
Ensemble (Ours)	Adult	54.8	7.84	37.37
	Adolescent	79.97	13	7.03
	Children	67.87	8.01	24.12

While short-term simulation does not imply uncertainties, their accumulation becomes inevitable in case of prolonged simulation. Namely, such factors as fluctuations of eating patterns, delayed glucose reactions, insulin stacking effect, etc., contribute to uncertainty. During all simulations, SAC and TD3 agents behaved unsteadily and provided extreme glucose peaks causing premature episode termination. Thus, analysis of the performance was carried out only for the proposed ensemble framework, whose behavior was stable and safe during all simulation duration.

4.2.2. Cohort-wise Performance in Long-Term Simulation

As can be seen from the table, the proposed model provides good results in terms of metrics in any cohort considered while certain deviations are possible because of physiological characteristics.

Specifically, adolescents' results demonstrate the highest value of Time in Range (79.97%) and lowest rates of hyperglycemia (7.03%). This fact proves the possibility of successful adaptation of the model to stable dynamics of glucose levels.

In the child cohort, the model is able to achieve a TIR of 67.87%, along with moderately low hypoglycemia of 8.01% and hyperglycemia of 24.12%. Even though there still exists high variability when compared to the adolescent group, the model succeeds in avoiding highly unsafe conditions, thus highlighting the effectiveness of the safety constraint approach for controlling the glucose levels.

For the adult group, the performance level of the model is somewhat lower, with TIR at 54.8% and high levels of hyperglycemia at 37.37%. These results suggest that long-term glucose regulation for adults requires more effort due to the delayed insulin release and high metabolic variation. However, even under such conditions, the controlled hypoglycemia of the model is at 7.84%.

To prove the proposed framework maintains its stability and robustness through time the researchers showed glucose patterns during the 10-day simulation in Figures 4.10–4.12 which included data from adult and adolescent and child study groups. The plots show the average glucose pattern together with the range of variation (±1 standard deviation) to help to understand both performance and consistency over long time periods.

The adolescent group keeps their glucose values inside the desired range because they show very little change which proves they have effective glucose management. The child group shows average changes because their bodies react more strongly but the system

manages to prevent unsafe conditions from becoming extended periods of danger. The adult group experiences more unpredictable blood sugar changes and their high blood sugar episodes become more frequent which makes it challenging to maintain stable glucose levels.

The meta-ensemble framework demonstrates stable glucose patterns throughout various time ranges when tested with real-world data which proves its ability to handle unpredictable situations.

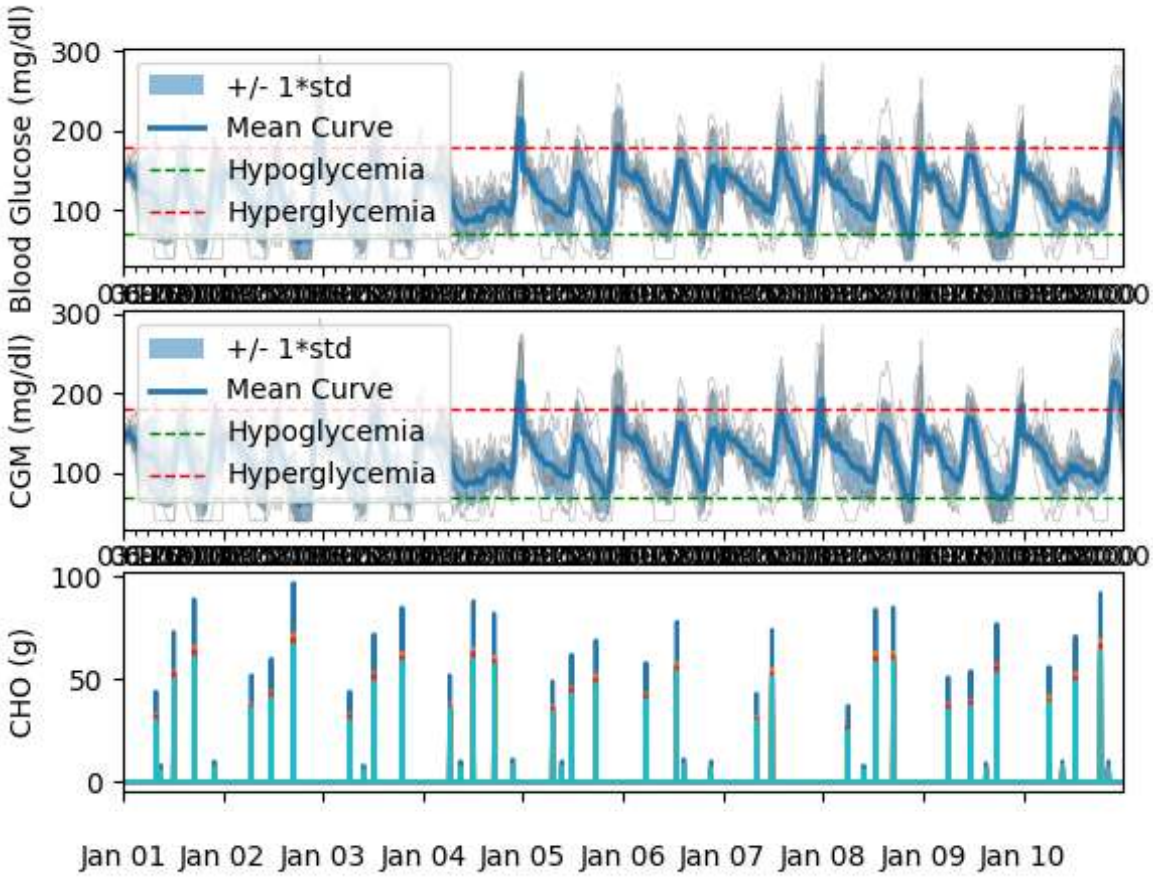

Figure 4.10. Long-Term Glucose Dynamics for Adolescent Cohort (10-Day Simulation)

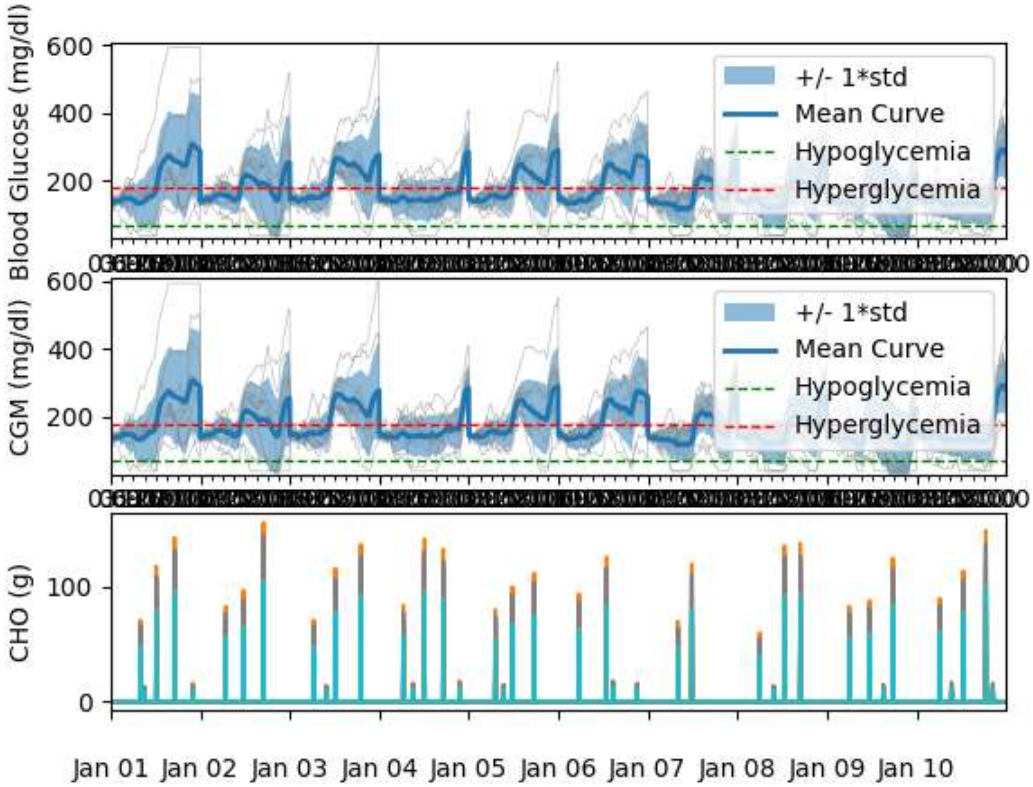

Figure 4.11. Long-Term Glucose Dynamics for Adult Cohort (10-Day Simulation)

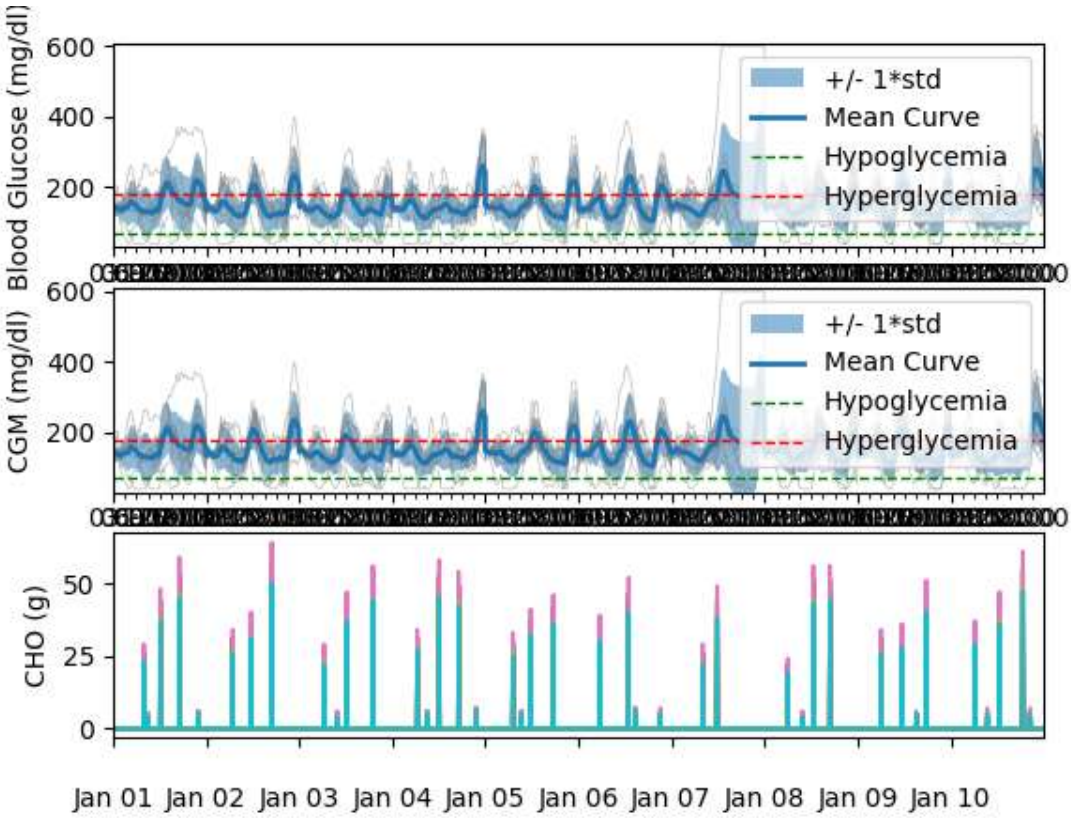

Figure 4.12. Long-Term Glucose Dynamics for child Cohort (10-Day Simulation)

4.5 Results Analysis

To further analyse the performance of the suggested framework on patients' outcomes and glucose zones can be seen for each cohort. They are provided in Tables 4.5-4.7 and Figures 4.13-4.15.

Table 4.5: Patient-wise Performance Metrics for Adolescent Cohort

Patient	TIR (%)	Hypo (%)	Sev. Hypo (%)	Hyper (%)	Mean BG (mg/dL)
adolescent#001	100	0	0	0	120.34
adolescent#002	74.05	1.04	0	24.91	160.37
adolescent#003	67.47	32.53	12.8	0	90.51
adolescent#004	89.27	0	0	10.73	131.75
adolescent#005	82.01	1.04	0	16.96	133.32
adolescent#006	92.04	1.38	0	6.57	132.72
adolescent#007	87.2	0	0	12.8	146.08
adolescent#008	69.2	0	0	30.8	157.06
adolescent#009	100	0	0	0	139.87
adolescent#010	67.82	17.99	12.46	14.19	128.25

As can be observed from Table 4.5 and Figure 4.13, adolescents have demonstrated the highest performance level among other cohorts. Most patients have shown high Time in Range with no hypoglycemic and hyperglycemic events, which means stable and uniform blood glucose levels throughout the day. The zone distribution figure supports the statement that time was spent in the target zone with no exposure to extreme levels of blood glucose.

Apart from individual-level analysis, it is worth noting that the overall performance statistics serve as additional evidence of the high efficiency of the developed model when applied to adolescents. In particular, the average Time in Range (TIR) is 82.91%, meaning that the majority of people keep their glucose levels within the clinically recommended boundaries. At the same time, the average incidence of hypoglycemia and severe hypoglycemia are rather low and are equal to 5.40% and 2.53%, respectively.

Thus, the designed algorithm manages to avoid hazardous low levels of blood glucose. Additionally, it is important to emphasize the mean hyperglycemia rate of 11.70% as a positive indicator because the value falls within acceptable limits. Finally, the mean level of blood glucose equals 134.03 mg/dL, which again stays within normal boundaries. It is

noteworthy that the adolescent group shows the highest performance rate among all analyzed groups.

Table 4.6: Patient-wise Performance Metrics for Adult Cohort

Patient	TIR (%)	Hypo (%)	Severe Hypo (%)	Hyper (%)	Mean BG
adult#001	100	0	0	0	120.2
adult#002	91.7	8.3	2.08	0	104.48
adult#003	84.08	0	0	15.92	159.66
adult#004	74.05	20.76	16.96	5.19	107.95
adult#005	83.04	0	0	16.96	148.18
adult#006	37.37	0	0	62.63	216.72
adult#007	60.21	15.57	5.54	24.22	138.72
adult#008	48.62	51.38	46.79	0	86.09
adult#009	52.25	0	0	47.75	192.2
adult#010	66.44	0	0	33.56	172.7

Moderate performance level with high variation was seen among adults, however Table 4.6 and Figure 4.14 demonstrate that some patients had high levels of hyperglycemia due to late glucose responses. Thus, the zone distribution figure proves that there was a clear tendency towards hyperglycemia among the cohort.

Apart from patient-specific data, the results of aggregate performance measures among the adult group reveal issues regarding glycemic management. The average TIR among the patients is reported as 68.72%, which seems quite low as compared to the results obtained by the adolescent group, thus suggesting that the group performs moderately. The average rate of hypoglycemia and severe hypoglycemia among adult participants is found to be 5.3% and 5.1%, respectively.

In terms of the average rate of hyperglycemia, it is found that this rate is quite high (25.99%), revealing a tendency of patients to suffer from hyperglycemia. This finding is consistent with individual patient-level data because some of them face considerable problems with hyperglycemia due to their late response to glucose. It is also important to mention that the mean blood glucose level is 144.69 mg/dL, which is slightly higher than the desirable value. Thus, it can be stated that the group of adults has a moderate level of performance.

Table 4.7: Patient-wise Performance Metrics for Child Cohort

Patient	TIR (%)	Hypo (%)	Sev. Hypo (%)	Hyper (%)	Mean BG (mg/dL)
child#001	48.79	43.94	31.49	7.27	90.44
child#002	55.36	37.72	25.26	6.92	94.02
child#003	82.01	2.77	0	15.22	115.05
child#004	47.75	40.14	31.49	12.11	100.24
child#005	77.51	11.42	3.46	11.07	132.16
child#006	85.12	6.57	2.42	8.3	127.47
child#007	81.31	0	0	18.69	140.91
child#008	53.63	32.18	23.18	14.19	114.4
child#009	69.9	5.19	1.04	24.91	145.5
child#010	58.48	1.73	0	39.79	187.03

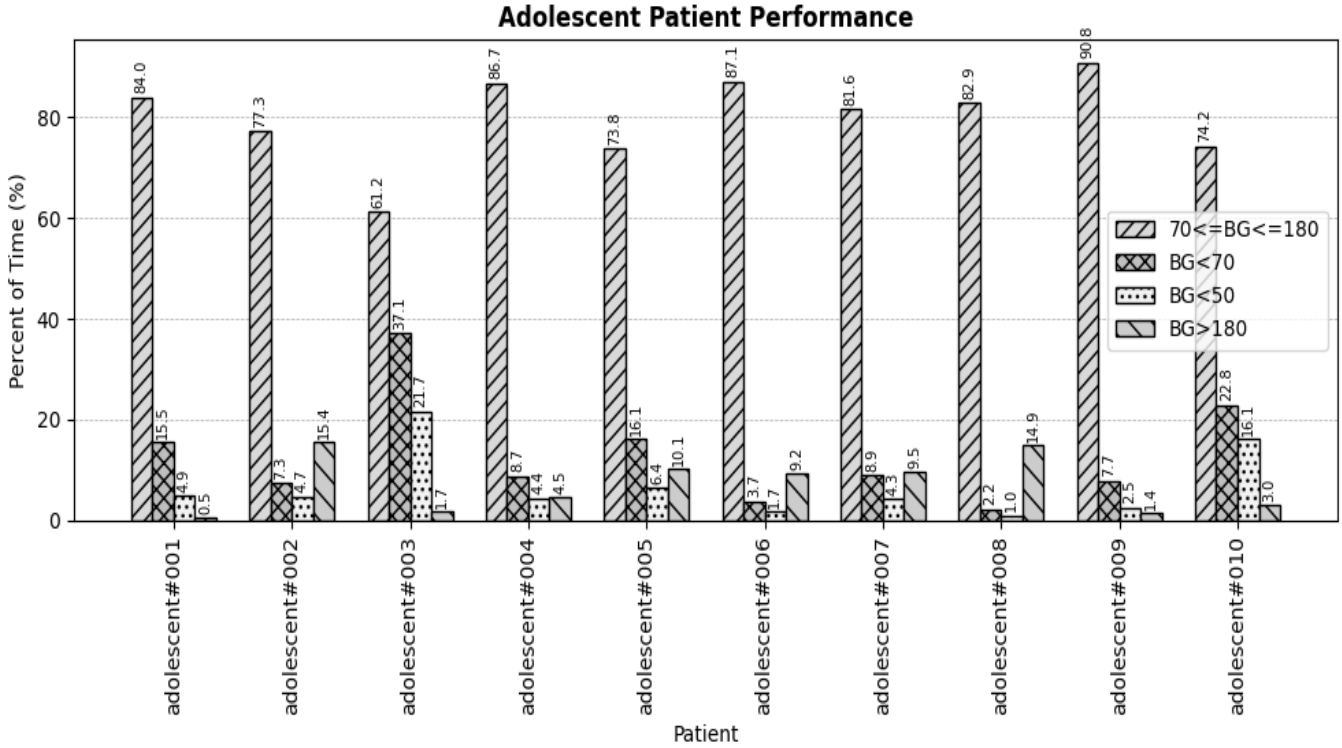

Figure 4.13. Patient-wise Glucose Zone Distribution for Adolescent Cohort

As shown in Table 4.7 and Figure 4.15, the children group presents average to good results along with higher variability. In other words, even though some patients achieve good Time in Range, others experience more frequent hypoglycemia, which includes severe hypoglycemia, due to their higher insulin sensitivity. This variability is also confirmed in the

zone distribution graph, which shows wider dispersion in glucose levels, especially in hypoglycemic zones.

Beyond individual patients' performances, the average performances for the group of pediatric patients reveal the presence of notable variations when it comes to blood glucose regulation. For instance, the average TIR is 65.99%. This value represents relatively moderate performance. Additionally, the mean values for hypoglycemia and severe hypoglycemia are 18.17% and 11.83% respectively, and they are much higher than those for other groups due to increased insulin sensitivity.

The average value for hyperglycemia is 15.85%, and its relative moderateness indicates that there are mechanisms by which the model can control high blood sugar values to some extent. In turn, the average value for blood glucose is 124.72 mg/dL, and it tends to be within the acceptable range. Overall, it is possible to conclude that the averaged data confirm the fact that there are more variations in the case of pediatric patients especially related to hypoglycemia.

In summary, the study reveals that the suggested model achieves its best performance on the adolescent group, has a satisfactory adaptation rate on the children group, and shows average results on the adults group due to physiology-related issues.

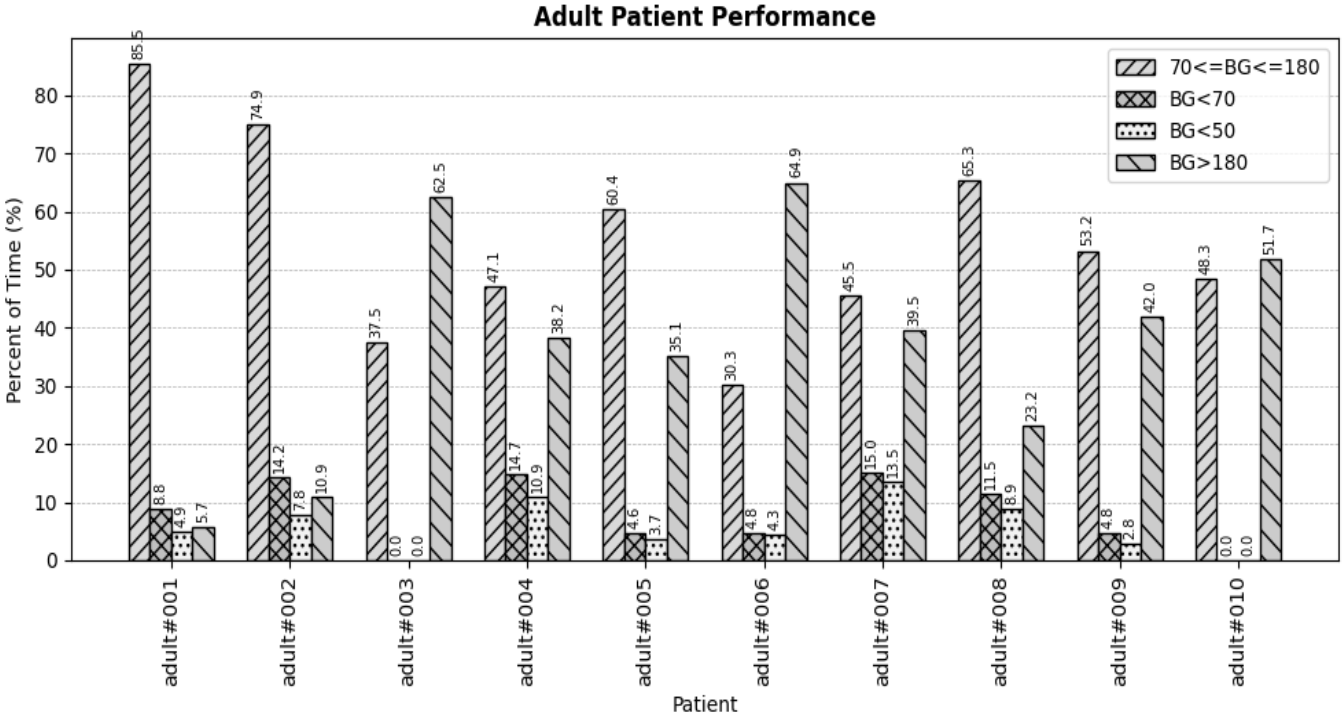

Figure 4.14. Patient-wise Glucose Zone Distribution for Adult Cohort

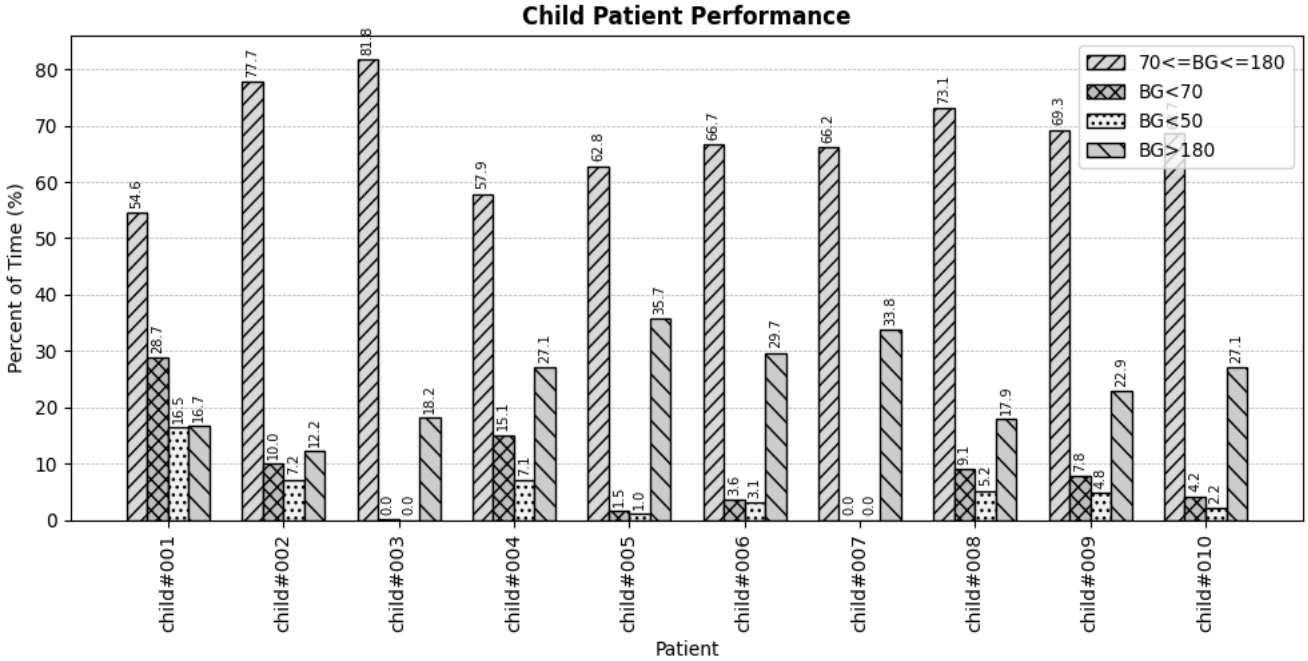

Figure 4.15. Patient-wise Glucose Zone Distribution for Child Cohort

4.6 Visualization of Glucose Control Behaviour

In order to analyze the qualitative performance of the framework, the glucose profiles along with the response insulin dosages are presented graphically for each of the selected patients from the two cohorts. These plots are provided in Figures 4.16–4.18.

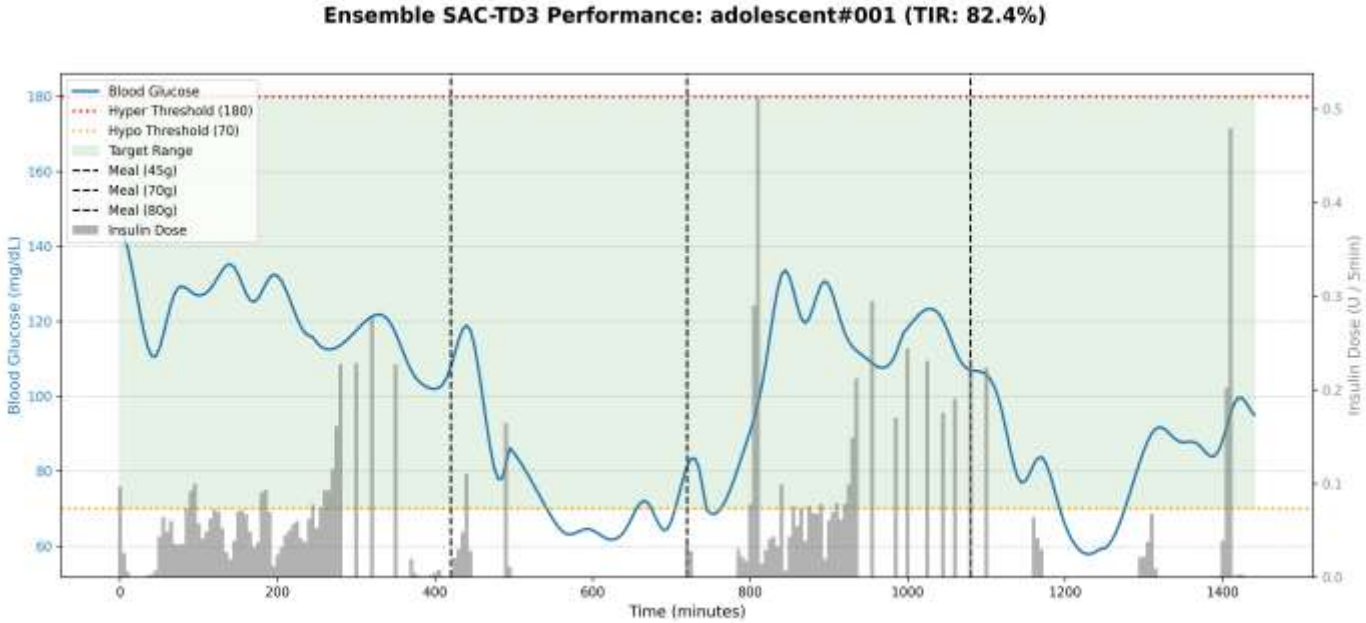

Figure 4.16 Glucose and insulin response of the Meta-Ensemble Agent for Adolescent

Figure 4.16 shows the closed-loop glucose control behavior of Meta Ensemble Agent for patient adolescent#001. The blood glucose level mostly stays within the clinically recommended target level of (70-180 mg/dL). Thus, a high TIR of 82.4% was achieved for this patient. Also, there is quick and clear response to meal disturbances for the glucose profile.

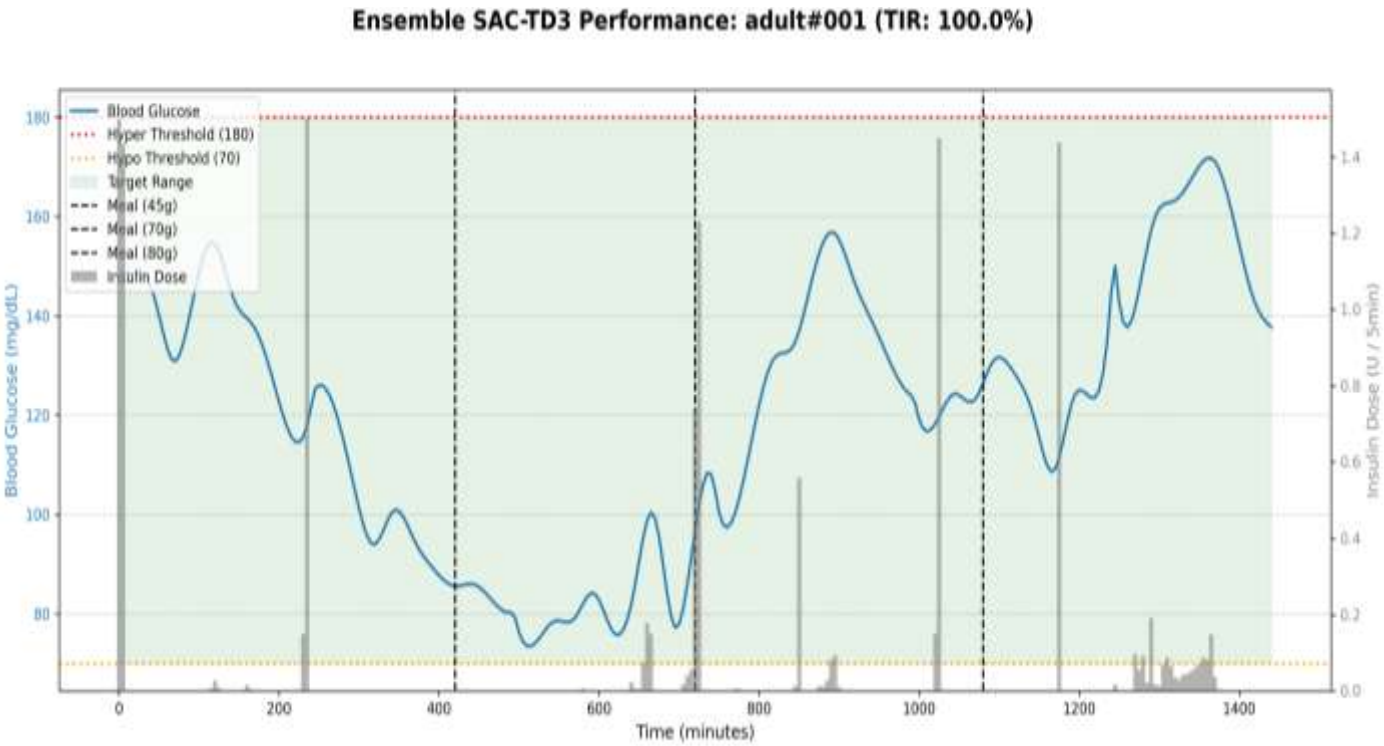

Figure 4.17 Glucose and insulin response of the Meta-Ensemble Agent for Adult

Figure 4.17 shows the closed-loop glucose control behavior of Meta Ensemble Agent for patient adult#001. The blood glucose level stays fully within the clinically recommended target level of (70-180 mg/dL). A complete TIR of 100% was achieved for this patient. Smooth response can be seen for the meal disturbances on the glucose profile.

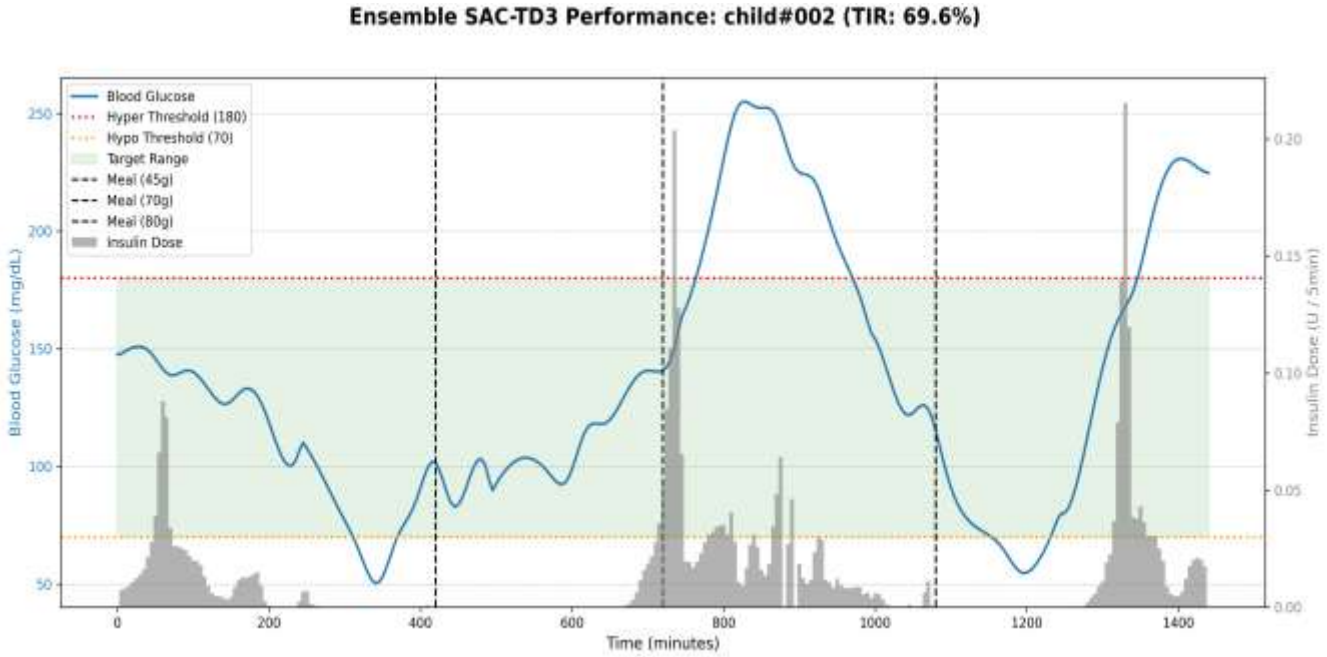

Figure 4.18 Glucose and insulin response of the Meta-Ensemble Agent for Child

Figure 4.18 shows the closed-loop glucose control performance of the Meta Ensemble Agent for child#002. While the graph is well within the target range for most of the time (TIR: 69.6%), some variation exists due to the meal disturbance problem. In terms of insulin dosage, the controller works appropriately to keep the level constant. However, there is hyperglycemia and hypoglycemia, indicating that control of the glucose levels is more difficult for the child group.

4.7 Observations from the Work

- The presented meta-ensemble reinforcement learning algorithm outperforms both SAC and TD3 agents consistently.
- For the adolescent cohort, the algorithm performs best and exhibits the highest time-in-range (TIR) score of 82.91% while at the same time ensuring relatively low hypoglycemia (5.4%) and relatively low hyperglycemia (11.7%).
- The child cohort shows decent adaptability and delivers an acceptable TIR of 65.99%. However, there is quite a lot of variability because of the increased insulin sensitivity and, therefore, higher levels of hypoglycemia (18.17%) in some patients.
- The adult group shows an average level of performance with TIR being equal to 68.72% due to the relatively high value of hyperglycemia (25.99%).

- The safety layer serves as an essential defense system which prevents dangerous insulin doses from reaching patients while it effectively protects all patient groups from hypoglycemia.
- The system operates through a closed-loop approach which does not need users to announce their meals because it demonstrates its ability to adjust automatically to blood sugar fluctuations while keeping blood sugar levels within a safe range.

CHAPTER-5
CONCLUSION AND FUTURE WORK

5.1 Conclusion

The current work offers the concept of a cohort-aware Meta-Ensemble Reinforcement Learning strategy aimed at delivering personalized, closed-loop diabetes therapy based on the use of SimGlucose. SAC and TD3 are used to form the core of the algorithm that will be capable of delivering personalized closed-loop insulin treatment based on patient physiological conditions.

An important novelty in terms of this research concerns the use of physiological parameters that play a key role in insulin sensitivity and blood glucose regulation. This aspect is associated with body weight, which is an important physiological feature influencing these processes. Furthermore, the use of a safety-constrained control system along with insulin-on-board regulation provides the ability to deliver personalized treatment that is considered clinically feasible.

According to experimental results, the proposed approach provides high glucose levels control. For instance, TIR of 82.91%, 68.72%, and 65.99% were reported by the model for adolescent, adult, and pediatric patient groups, respectively. Hypoglycemia rates remained low, and the algorithm did not lead to a significant occurrence of this condition in all groups. Compared to other methods, SAC and TD3-based systems demonstrated better stability and effectiveness. Importantly, the presented approach does not rely on meal announcement.

The combination of reinforcement learning algorithms and clinical safety rules effectively solves both gradual changes and abrupt events, such as meals' consumption. This capability is highly important for any application in medicine, in which consistent decision-making is crucial.

An additional advantage demonstrated by the proposed framework can be highlighted from the analysis of patients' Time-in-Range and Zone Distribution. Namely, the framework demonstrates excellent performance on individual patients despite the differences in their physiology. In the case of adolescents, a majority of the patients, specifically around 8-9

patients, show a substantial percentage of Time in Range in the desirable range of 70-180 mg/dL. For the Children group, the framework allows up to 6-7 patients to achieve an acceptable result in terms of Time-in-Range and a limited number of hypoglycemia events. However, there is a considerable variation in terms of patients' response that results from higher insulin sensitivity. Adults show relatively mediocre results compared to other patient cohorts, as only about four to five people stay in the target interval. Moreover, some cases demonstrate an increasing number of hyperglycemic episodes in adults.

Despite these differences, it can be claimed that the model demonstrates generalizable performance while accounting for specific traits of each patient. The framework successfully combines the benefits of reinforcement learning with safety guarantees provided by a set of medical rules. Furthermore, the ability to consider patient type makes the suggested solution suitable for various groups of patients. In particular, the inclusion of different patient groups allows capturing the difference in physiological traits and insulin sensitivity. It, in turn, increases the overall robustness and reduces the necessity for re-training the model for each new patient.

To conclude, the developed meta-ensemble approach is a good foundation for creating personalized and adaptive control strategies for automating the insulin delivery process.

5.2 Advantages of the Proposed System

The meta-ensemble reinforcement learning framework which has been proposed brings multiple benefits to diabetes management when compared to standard methods and single-agent systems. The system achieves improved glucose control through a meta-controller which merges SAC with TD3 to maintain equilibrium between exploration activities and stable system behavior. The framework achieves strong generalization when tested on different patient groups which include adults and adolescents and children which leads to better application potential. The system achieves better personalization through its use of body weight and other physical characteristics which allows it to generate insulin doses that match each person's needs.

The system achieves clinical reliability through its safety-based control mechanism which blocks dangerous insulin dosages and decreases the chance of dangerous low blood sugar episodes. The system operates with enhanced security features which makes it perfect for actual implementation in environments that demand patient protection as their main concern.

The proposed framework operates through a unique system which does not need users to announce their meals or provide any information about their carbohydrate consumption. The model automatically detects glucose changes which drive its adaptive response system to deliver insulin through automated operation. The system decreases user workload through this feature which makes it simple for people to use the system for diabetes management that requires continuous monitoring. The proposed method offers an automated insulin delivery system which adapts to various conditions and scales well for clinical use in medical environments.

5.3 Limitations of the Study

The proposed framework shows excellent results when tested through simulations but the current research evaluation remains limited to the SimGlucose environment. The research community uses this clinically validated simulator for artificial pancreas system studies yet it fails to represent all physiological changes which humans experience along with the unpredictable nature of actual human body responses. The simulation process simplifies three major elements which include sensor noise and patient-specific behavioral patterns and unexpected disturbances that could affect actual system functionality.

The existing framework employs a concise state representation which unites essential physiological indicators through blood glucose patterns and insulin-on-board and body weight measurements. The system fails to represent three essential elements which include physical exercise and stress responses and different food intake habits. The system would respond better to environmental changes when scientists introduce these elements because they create complex and changing effects on glucose metabolism.

The proposed model shows effective generalization across various patient groups but the current version lacks the ability to adapt its functions for specific patients during their online use. The framework operates through cohort-based learning which enables generalization across different groups yet future development should include patient-specific tuning systems to enhance individual results while keeping system stability intact.

5.4 Future Study

Further research is needed to expand the applicability and increase the safety of the Meta Ensemble (SAC + TD3) approach to enable a wider use in different patient groups. Although the current model shows good results in simulation, additional developments are needed to

apply the system effectively in reality. One of the ways to enhance the performance is the customization of the ensemble policy by adapting it individually to patients depending on their physiological characteristics (e.g., the level of insulin sensitivity and daily routine). The process can include tuning of the meta-controller or using an online learning approach to adjust the parameters in real time.

To increase the safety of the policy, the model can be improved by applying recurrent models to represent long-term glucose levels and delayed effects of insulin administration. It is possible to refine the reward function of the system by adding clinically proven risk metrics. Also, it would be useful to evaluate the system in a variety of scenarios with diverse factors like meal schedules, physical activity, or other factors that affect blood glucose levels.

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APPENDIX

```python
class MetaController(nn.Module):
    """
    A small neural network that looks at the physiological state and
    outputs the optimal weights for the SAC and TD3 agents.
    """
    def __init__(self, state_dim):
        super(MetaController, self).__init__()
        self.net = nn.Sequential(
            nn.Linear(state_dim, 64),
            nn.ReLU(),
            nn.Linear(64, 2),
            nn.Softmax(dim=-1) # Ensures weights sum to 1.0
        )

    def forward(self, state):
        raw_weights = self.net(state)
        # THE SAFETY CLAMP:
        # Squeeze the [0.0, 1.0] output into a [0.2, 0.8] range.
        # This guarantees the agent ALWAYS uses at least 20% of the other network,
        # preventing it from finding a fatal "local optima" of 100% SAC.

        # clamped_weights = 0.2 + (raw_weights * 0.6)
        clamped_weights = 0.1 + (raw_weights * 0.8)
        return clamped_weights
```

Figure A.1: Meta-Controller Network for Adaptive Weight Generation

```python
class EnsembleAgent:
    """
    State-of-the-art Trainable Ensemble Agent.
    Dynamically learns how to mix SAC and TD3 policies using a Meta-Controller.
    """
    def __init__(self, state_dim, action_dim, max_action, device, ...

    def select_action(self, state, evaluate=False):
        """
        Uses the Meta-Controller to dynamically weight the actions.
        """
        state_tensor = torch.FloatTensor(state).unsqueeze(0).to(self.device)

        # 1. Get base actions
        sac_action = self.sac_agent.select_action(state, evaluate=evaluate)
        td3_action = self.td3_agent.select_action(state, evaluate=evaluate)

        # 2. Get learned weights from Meta-Controller
        with torch.no_grad():
            weights = self.meta_controller(state_tensor).cpu().numpy()[0]

        w_sac, w_td3 = weights[0], weights[1]

        # 3. Combine actions
        action = (w_sac * sac_action) + (w_td3 * td3_action)

        return action, w_sac, w_td3
```

Figure A.2: Action Selection using Meta-Ensemble Strategy

```python
class SafetyLayer:
    def __init__(self, cohort='adult', hypo_threshold=80, hyper_threshold=180): ...

    def apply(self, action, state):
        glucose, rate_of_change, iob, bw = state

        # 1. HARD HYPO CUTOFF
        if glucose < self.hypo_threshold:
            return np.array([0.0])

        # 2. PREDICTIVE SUSPENSION (The "Control-IQ" Fix)
        # rate_of_change is roughly mg/dL per minute.
        # If our predicted glucose in 20 minutes is below 75, STOP insulin!
        future_bg = glucose + (rate_of_change * 20.0)
        if future_bg < 80.0:
            return np.array([0.0])

        # 3. IOB STACKING CUTOFF
        if iob >= self.max_safe_iob:
            return np.array([0.0])

        # 4. GENTLE BRAKES (If dropping fast, cut dose in half)
        if rate_of_change < -1.5:
            return np.clip(0.5 * action, 0, float('inf'))

        return action
```

Figure A.3: Safety-Constrained Control Layer for Insulin Delivery

```python
def compute_reward(g, rate, iob):
    reward = 0

    # Target zone reward (Gaussian shaping)
    if 80 <= g <= 140:
        reward += 20 * np.exp(-0.5 * ((g - 110) / 25) ** 2)

    # Hypoglycemia penalty (high priority)
    if g < 70:
        reward -= 250 * (1 + (70 - g) / 10)
    elif g < 54:
        reward -= 500 * (1 + (54 - g) / 10)

    # Hyperglycemia penalty
    if g > 180:
        reward -= min((g - 180) * 0.4, 80)

    # Stability bonus
    if abs(rate) < 1.5 and 70 <= g <= 180:
        reward += 10

    # IOB penalty (avoid insulin stacking)
    if iob > 10:
        reward -= (iob - 10) ** 1.5

    return np.clip(reward, -500, 100)
```

Figure A.4: Multi-Zone Reward Function for Glucose Regulation

```python
class ReplayBuffer:
    def __init__(self, capacity):
        self.capacity = capacity
        self.buffer = collections.deque(maxlen=capacity)

    def push(self, state, action, reward, next_state, done):
        self.buffer.append((state, action, reward, next_state, done))

    def sample(self, batch_size):
        state, action, reward, next_state, done = zip(*[self.buffer[i] for i in np.random.choice(len(self.buffer), batch_size)])
        return np.array(state), np.array(action), np.array(reward), np.array(next_state), np.array(done)

    def __len__(self):
        return len(self.buffer)
```

Figure A.5: Experience Replay Buffer for Reinforcement Learning

```python
class RealisticMealScenario(Scenario):
    """
    A realistic meal scenario generator based on the schema from Wang et al. [60]
    as described in Biomedicines 2024, 12, 2143.
    """
    def __init__(self, start_time, patient: T1DPatient, seed=None):

    def _generate_daily_meals(self):
        # (This internal logic is correct and remains unchanged)
        p = [0.95, 0.3, 0.95, 0.3, 0.95, 0.3]
        lo = np.array([5, 9, 10, 14, 16, 20]) * 60
        up = np.array([9, 10, 14, 16, 20, 23]) * 60
        mu_t = np.array([7, 9.5, 12, 15, 18, 21.5]) * 60
        sigma_t = np.array([60, 30, 60, 30, 60, 30])

        bw = self.patient._params['BW']
        mu_a_factors = np.array([0.7, 0.15, 1.1, 0.15, 1.25, 0.15])
        mu_a = mu_a_factors * bw
        sigma_a = mu_a * 0.15

        generated_meals = []
        for k in range(6):
            if self.random_gen.rand() <= p[k]:
                amount = self.random_gen.normal(mu_a[k], sigma_a[k])
                amount = round(max(0, amount))
                a, b = (lo[k] - mu_t[k]) / sigma_t[k], (up[k] - mu_t[k]) / sigma_t[k]
                time = truncnorm.rvs(a, b, loc=mu_t[k], scale=sigma_t[k], random_state=self.random_gen)
                time = round(time)
                if amount > 0:
                    generated_meals.append((time, amount))

        generated_meals.sort(key=lambda x: x[0])
        return generated_meals
```

Figure A.6: Stochastic Meal Generation based on Physiological Model

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